

ANALOG & DIGITAL CIRCUITS

Lecture 2

Computer Science Department, TUCN

The switching regime of the semiconductor devices

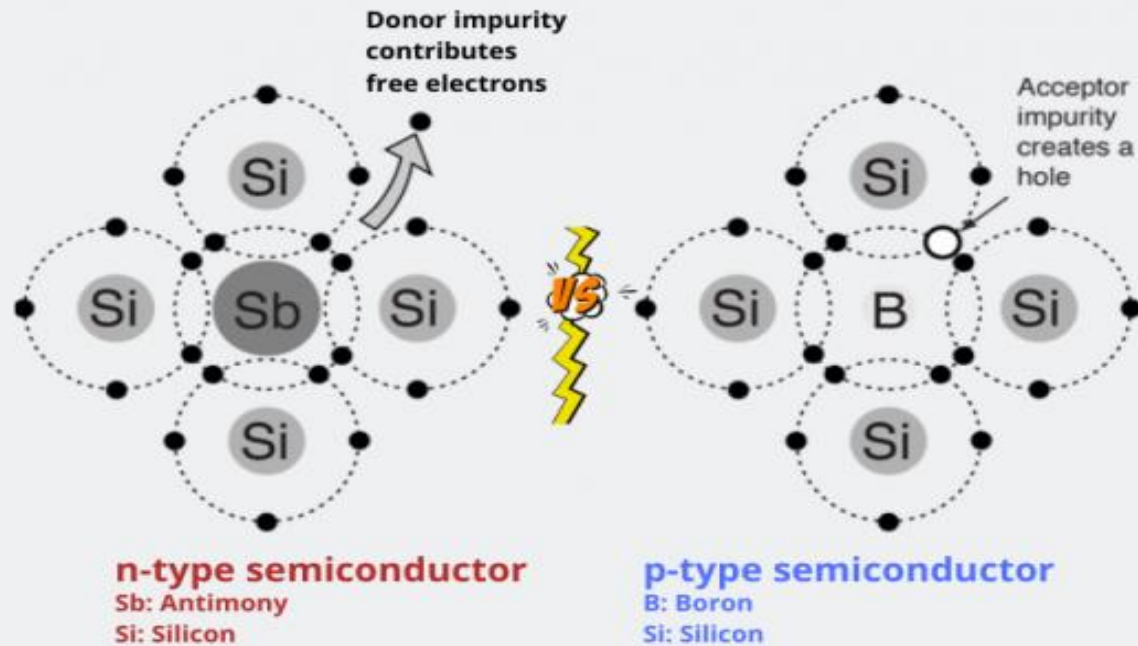
Diode

Bipolar Transistor

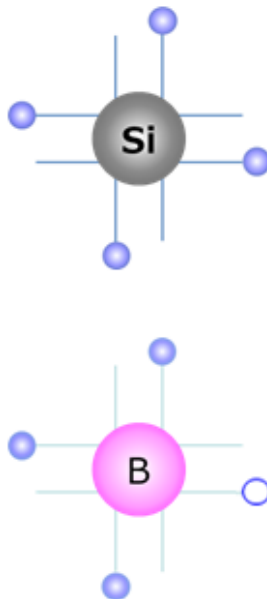
Unipolar MOS Transistor

Semiconductor Materials

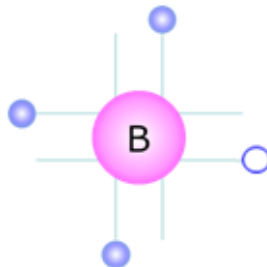
- Semiconductors: solid or liquid materials with an electric conductivity somewhere between the conductors and the insulators
 - conductors (metals): resistivity varies between 10^{-6} - 10^{-4} ohms-cm
 - insulator's resistivity (diamond, quartz) is between 10^{10} - 10^{20} ohms-cm
 - semiconductor materials (silicon and germanium): intermediary resistivities (hundreds or thousands ohms-cm); found on the 4th column of the Mendeleev table, having 4 valence electrons
- Semiconductor types:
 - **n type**, where the electrons are in excess; obtained by adding impurities like Phosphorus, Arsenic, elements that are found on the fifth column
 - **p type**, where charge carriers in excess are the positive ones (holes); obtained by adding impurities like Boron, Gallium or Aluminum, placed on the third column and having a three valence electrons



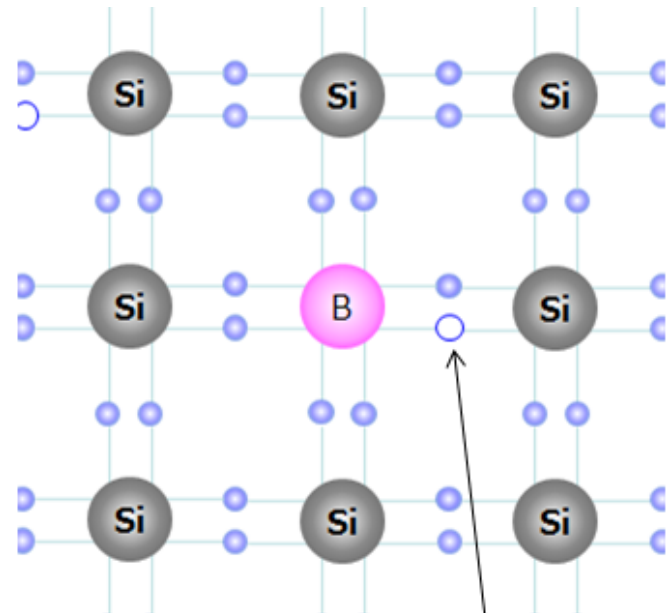
Silicon (Si):
Four valence
electrons



Boron (B):
Three valence
electrons

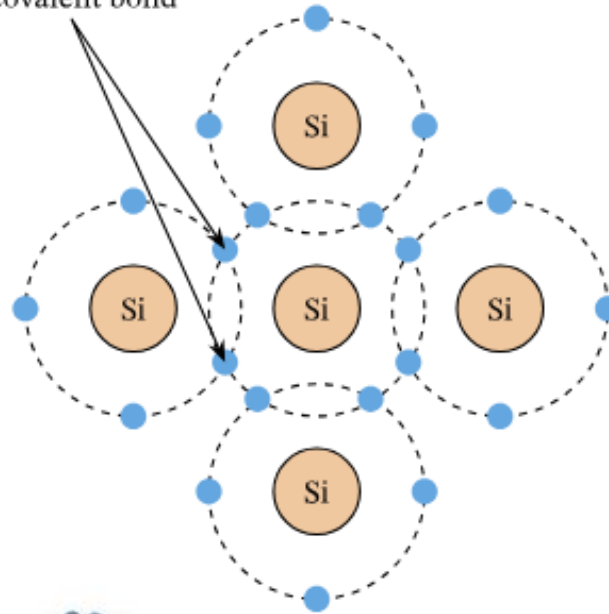


Adding boron to
pure silicon crystal
results in lack of an
electron. And it
becomes a hole.

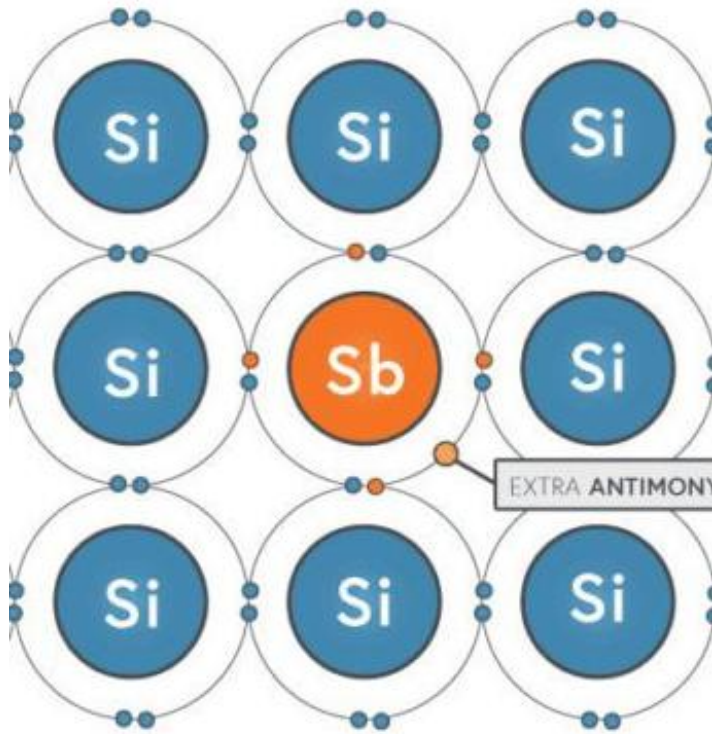


Hole

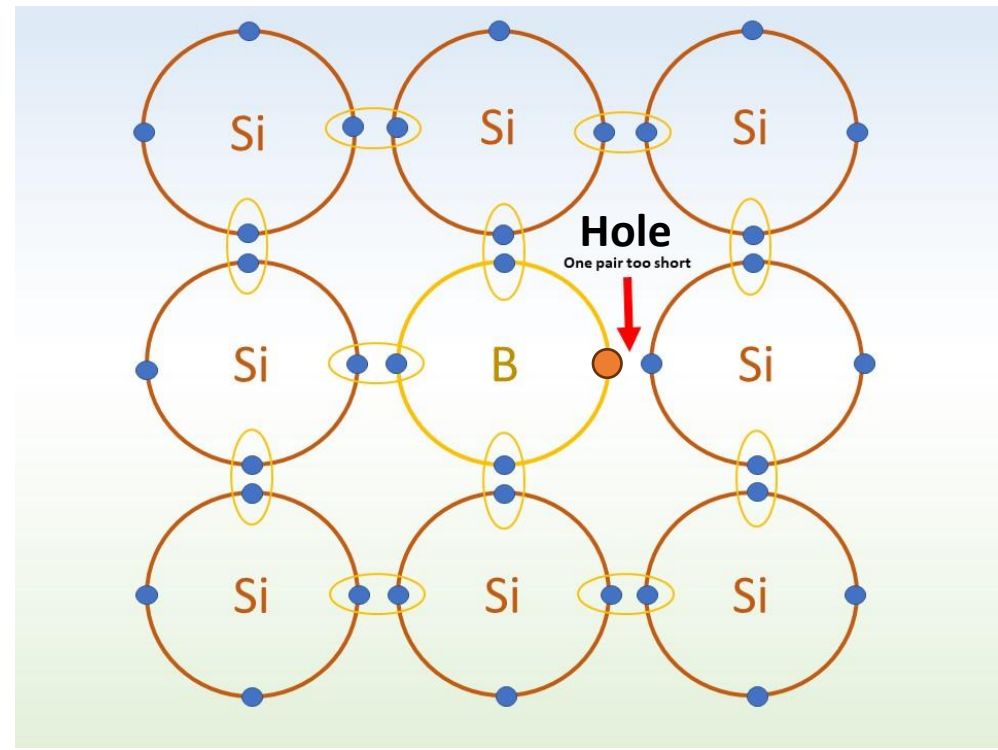
Shared electrons of
a covalent bond



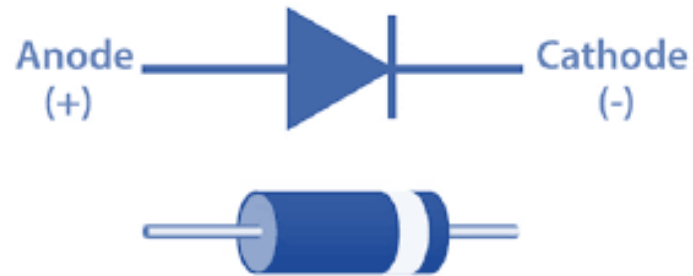
N-type lattice



P-type lattice



Diodes



Gunn Diode



LED

PIN Diode



Step Recovery Diode



Laser Diode



Photo Diode



Tunnel Diode



Varactor Diode



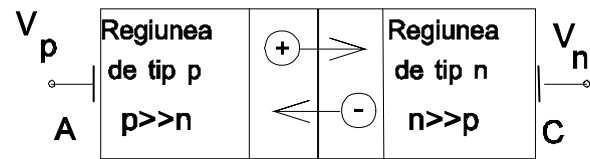
Schottky Diode



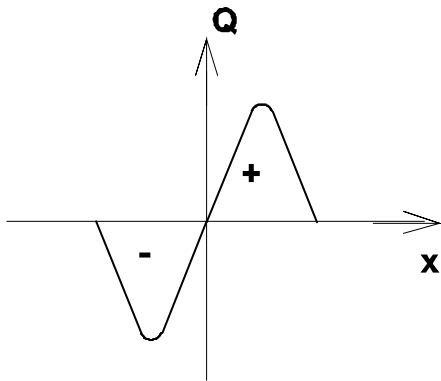
Zener Diode

Switching regime of the semiconductor diode

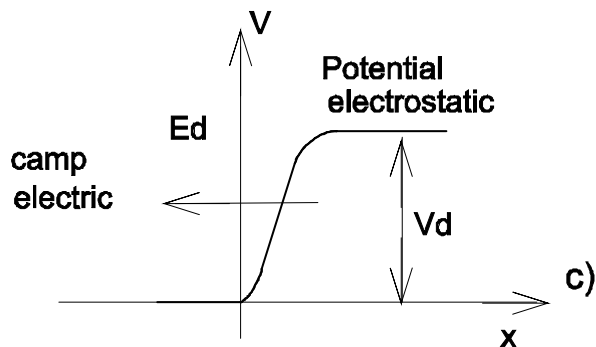
The pn junction at thermal equilibrium



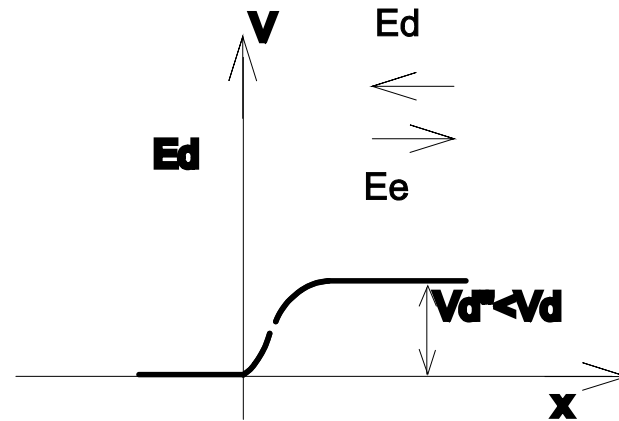
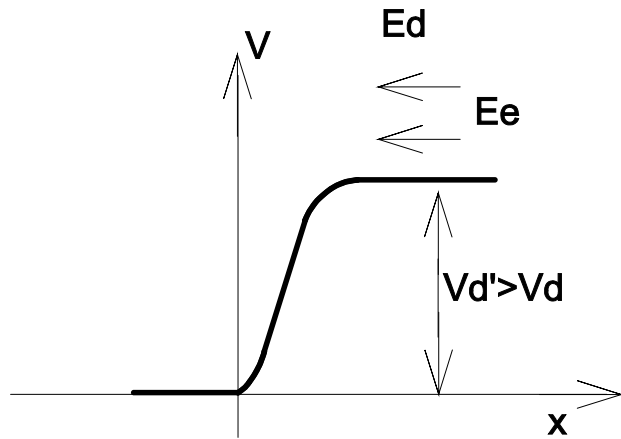
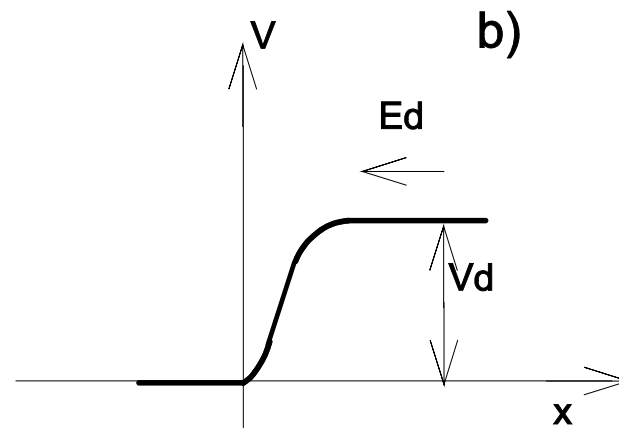
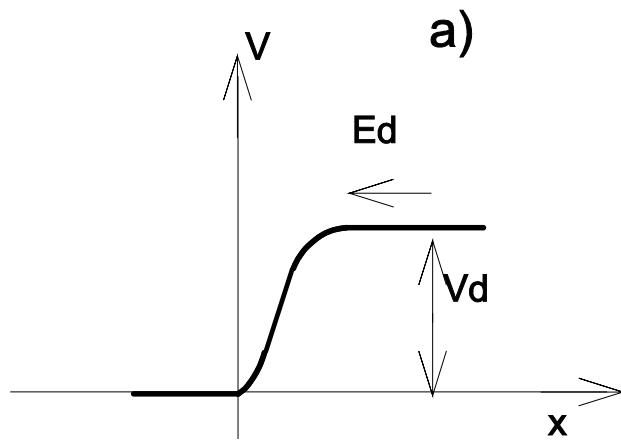
- a)
- Thermal equilibrium: no electrical current shall flow through the semiconductor
 - An organized movement of the charge carriers:
 - applying an external electrical field
 - inducing a non-uniform distribution of the charge carriers (diffusion process)



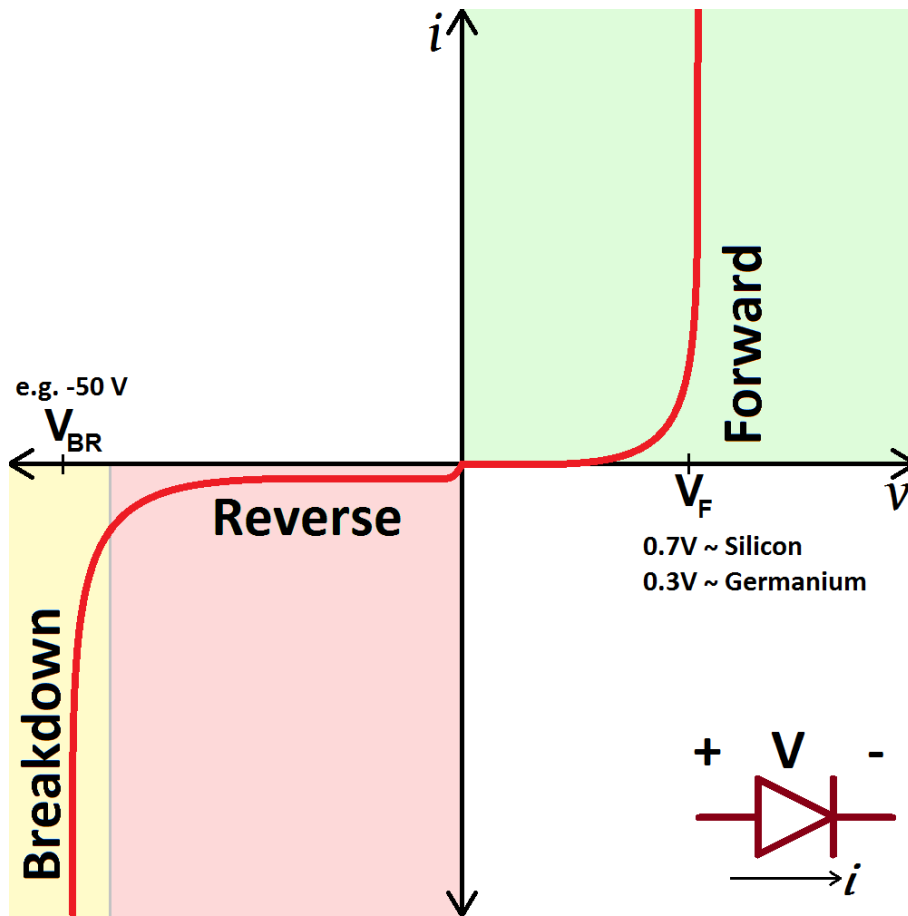
- b)
- Near junction plane, electrons from n-region tend to diffuse in p-region and similar for holes from p-region towards n-region



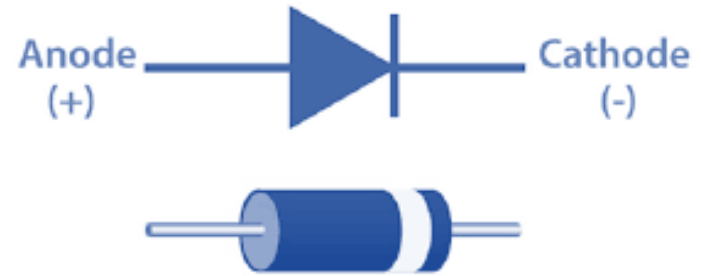
- Diffusion electric field (E_d): associated to a difference of potential, named *barrier potential* near the plane of the junction (due to presence of electrons in p-region and holes in n-region; opposes to the tendency of diffusion, so have a process with self-limitation)



- A voltage which increases the potential barrier is called **reverse polarity voltage**
- An external voltage of an opposite polarity is called **direct voltage**



Current-voltage characteristics



- Be a voltage applied on diode electrodes:

$$U = V_p - V_n$$
- **$V_p > V_n$** – great diode current due to the majority carriers;
 - the junction is **forward biased**
- **$V_p < V_n$** – current is negligible due to the minority carriers;
 - the junction is **reversely biased**

Static switching parameters of the semiconductor diode

For an ideal **pn** junction, the current-voltage relation is:

$$I = I_0(e^{U/c_r U_T} - 1)$$

I_0 is the reverse saturation current of the diode $I_0 = eS \frac{c_{dg} \cdot p_{m0}}{w}$
where:

e: electrical charge of the electrons ($e=1,6 \cdot 10^{-19} \text{C}$)

S: transversal section through the junction

c_{dg} : the diffusion coefficient for holes

p_{m0} : the concentration of the minority carriers

w: width of the recombining zone

c_r - recombining factor of the holes (has value 1 for Ge, value 2 for Si)

U_T is the thermal voltage:

$$U_T = \frac{KT}{e} = \frac{T}{11800}$$

K is the Boltzmann constant ($K= 1,38 \cdot 10^{-23} \text{J/}^\circ\text{K}$)

T is the absolute temperature

Static (dc) resistance of the diode, value depends on the functioning point

$$R_{cc} = \frac{U}{I}$$

Dynamic resistance (a.k.a. ac, differential)

$$R_d = \frac{\Delta U}{\Delta I}$$

Barrier capacity Cb, it depends nonlinear with the voltage

Diffusion capacity Cd, results from the stored charges through the diffusion of the minority carriers. It depends on the forward bias voltage

2 lines:

$i=0, u < U_T$

$i=u/R_d, u > U_T$

U_T threshold voltage,
 R_d differential resistance,,
 I_{D0} residual current

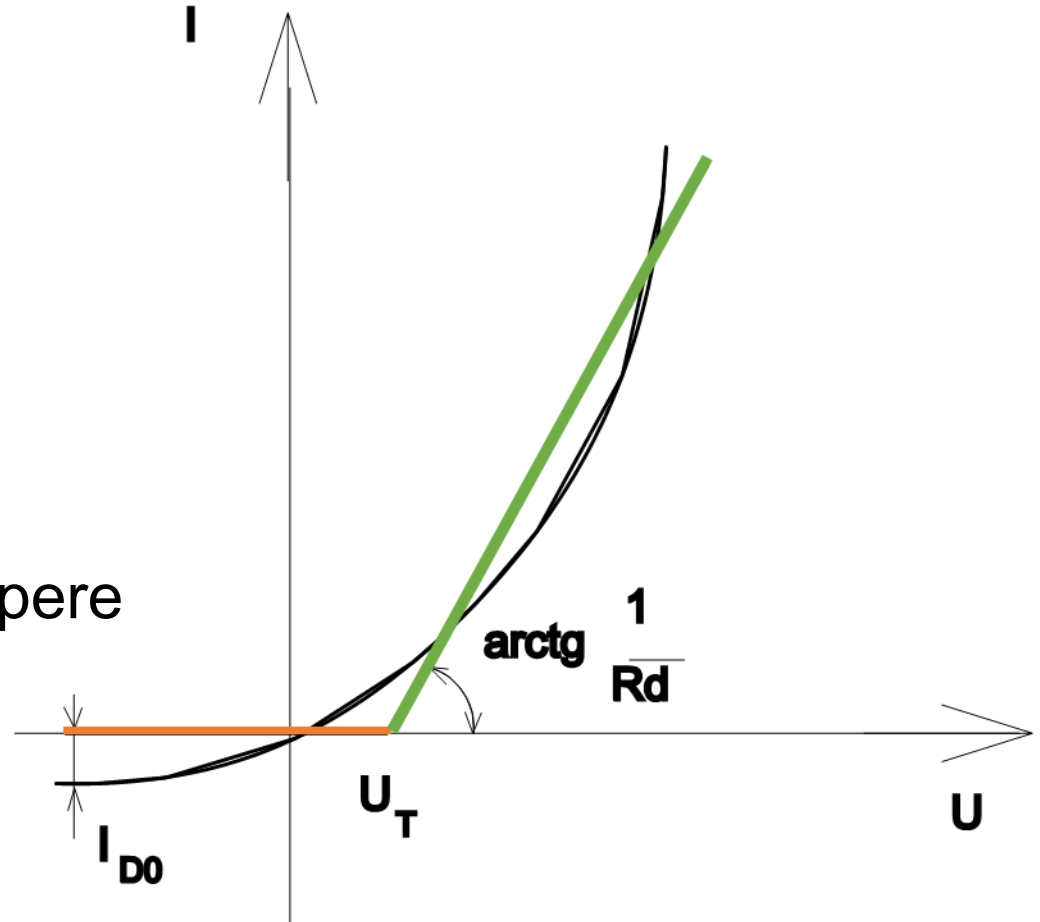
=> are giving the linear
approximation of the volt-ampere
diode characteristics

Typical values::

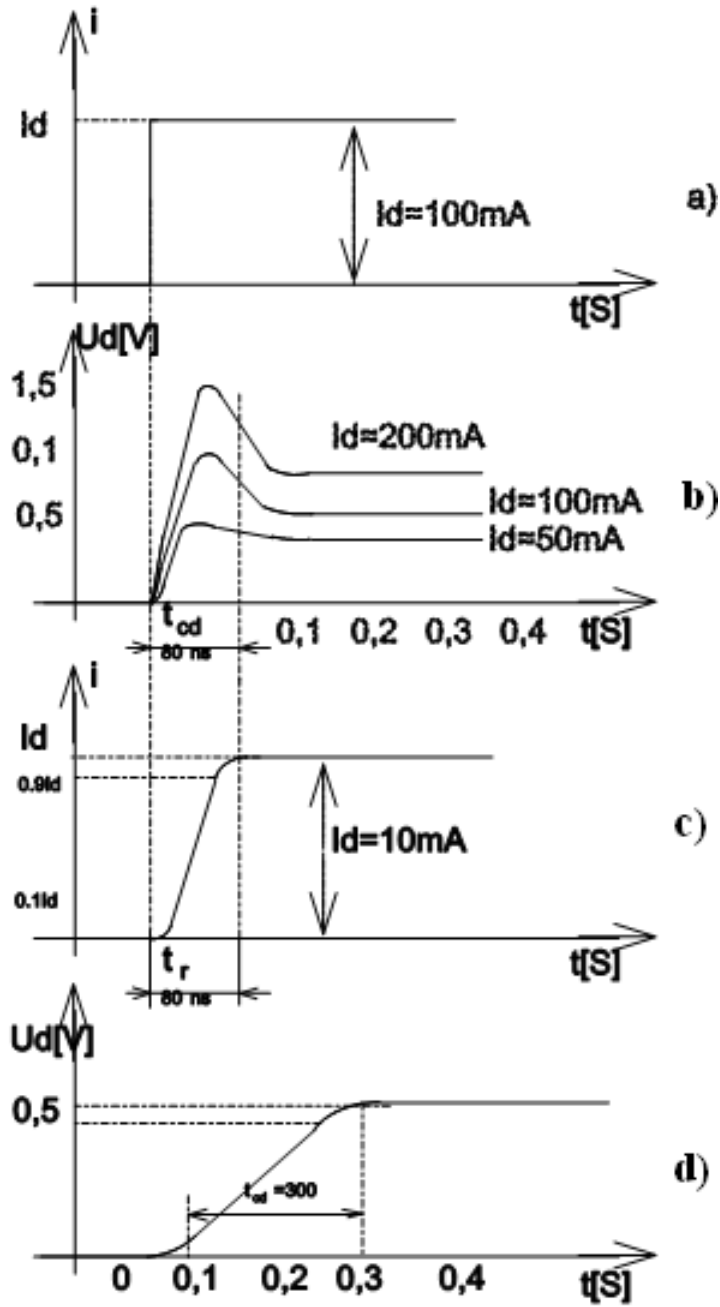
$I_0 = 0,05A$

$U_T = 0,5V$

$R_d = 15ohm$



Dynamic parameters of the semiconductor diodes



Forward switching time

- Necessary time for the diodes to pass from the blocked (off) state to the conduction state
- Smaller than the reverse switching (recovery) time

Two cases:

- the current pulse has a large amplitude and the anterior front is short; diode overvoltage (b)
- if the current pulse has a small amplitude and a less steep anterior front; the diode can be represented as a low-pass RC circuit (d)

Reverse switching time

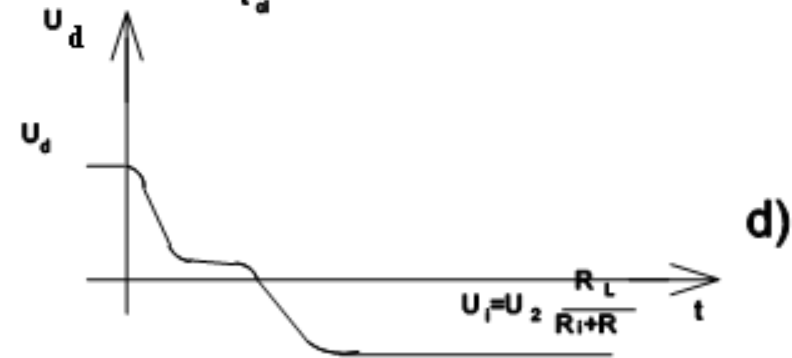
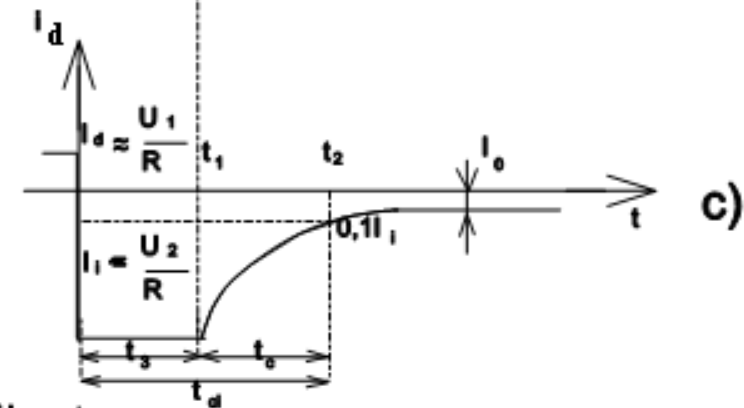
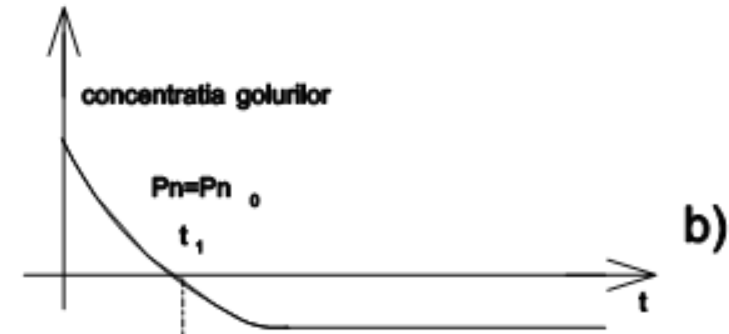
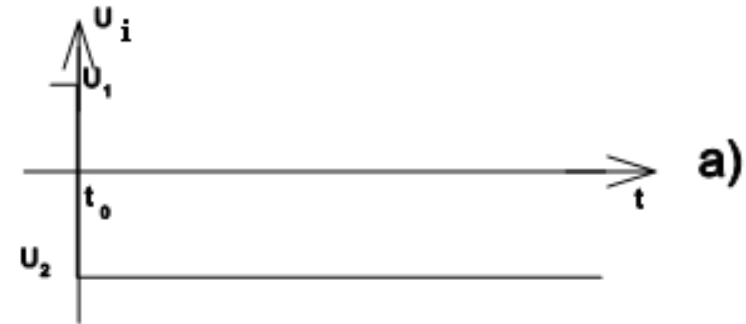
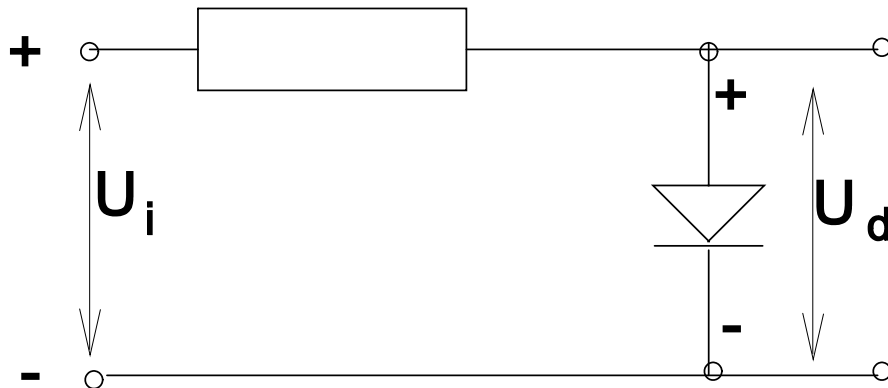
- Time when diode is passing from the conducting state to the blocking (off) state
- Switch analysis using circuit below

Two Time components:

- Storage time: $t_s = \tau \ln\left(1 + \frac{I_D}{I_I}\right)$
- Fall down time: $t_c = 2,3 \cdot R \cdot C_b$

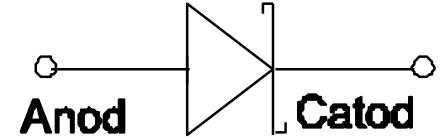
C_b – Barrier capacity of the diode

Reverse switching time: $t_{ci} = t_c + t_s$



Schottky barrier diodes (hot-carrier diodes)

- Based on a metal-semiconductor contact
- Electrical current is created by the movement of the majority carriers
- Manufactured from a semiconductor material (silicon) of **type n**, in contact with a metal, gold or Aluminum



Forward biasing of the metal: a junction current is created: movement through the junction of the electrons within the semiconductor

The current is created from the movement of electrons, the majority carriers have a greater energy than the one of the free electrons, name of hot carriers.

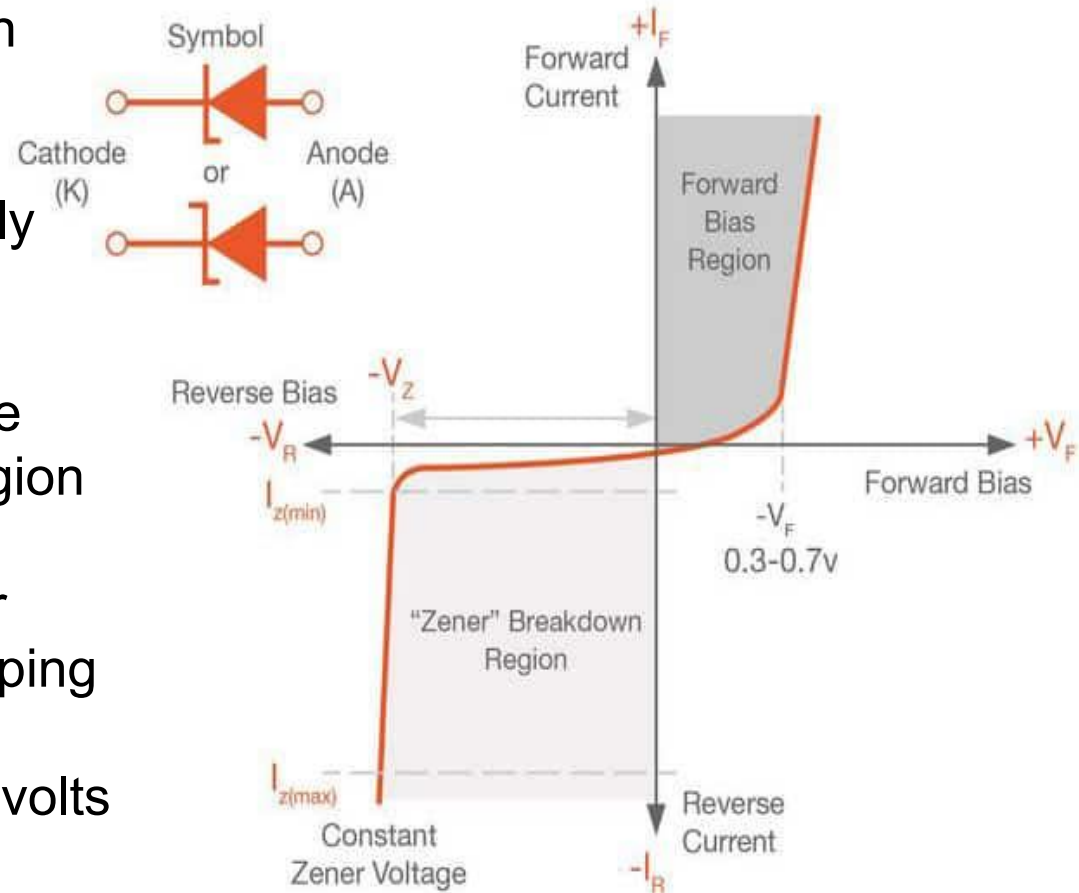
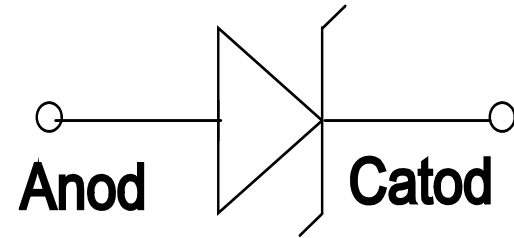
Reverse bias: electrons that broke through the metal are not distinguished from the free electrons of the metal; no residual current

Reverse switching time of the Schottky diode very small ($\approx 10\text{ps}$)

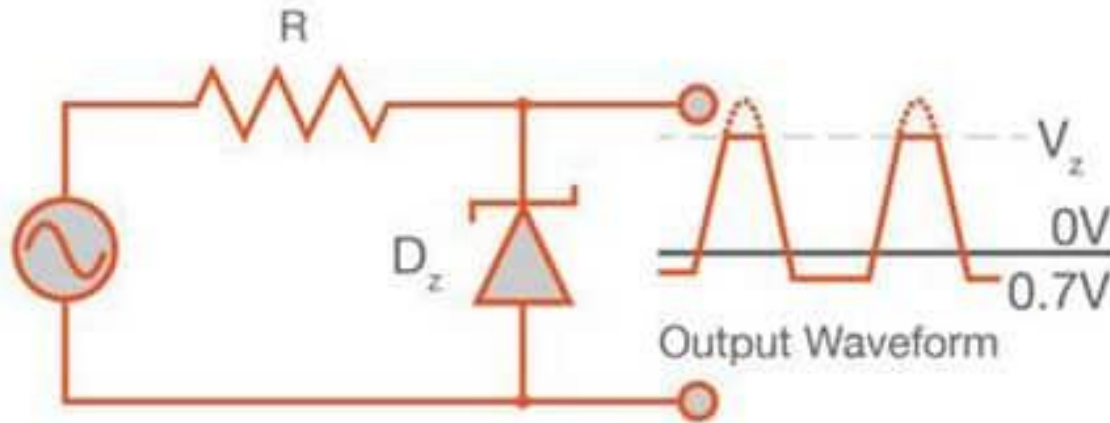
- Schottky diodes – used to create high-speed switching circuits
- The forward voltage fall on a Schottky diode is $\approx 0,4\text{V}$, unlike the $0,75\text{V}$ for the Si based diodes and the $0,3\text{V}$ for the Ge ones

Zener diodes

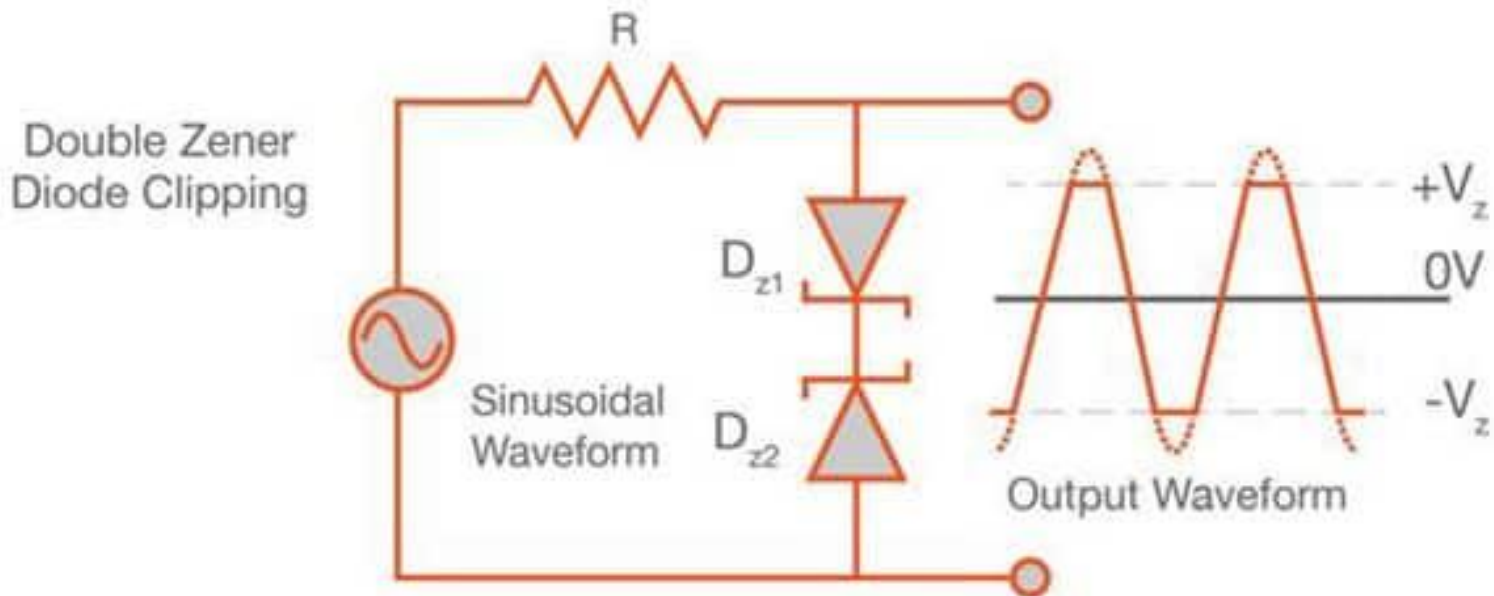
- Reverse biasing applied on a **pn** junction: if the existing electric field possesses the necessary energy to extract the electrons from their covalent rotating orbits, there will be a sudden increase of current
- The electron-hole pairs newly created contribute to the generation of an important current, and it is said that the diodes work in the Zener region
- Value of the threshold Zener voltage depends on the doping degree and can range from values of 2V to hundreds of volts



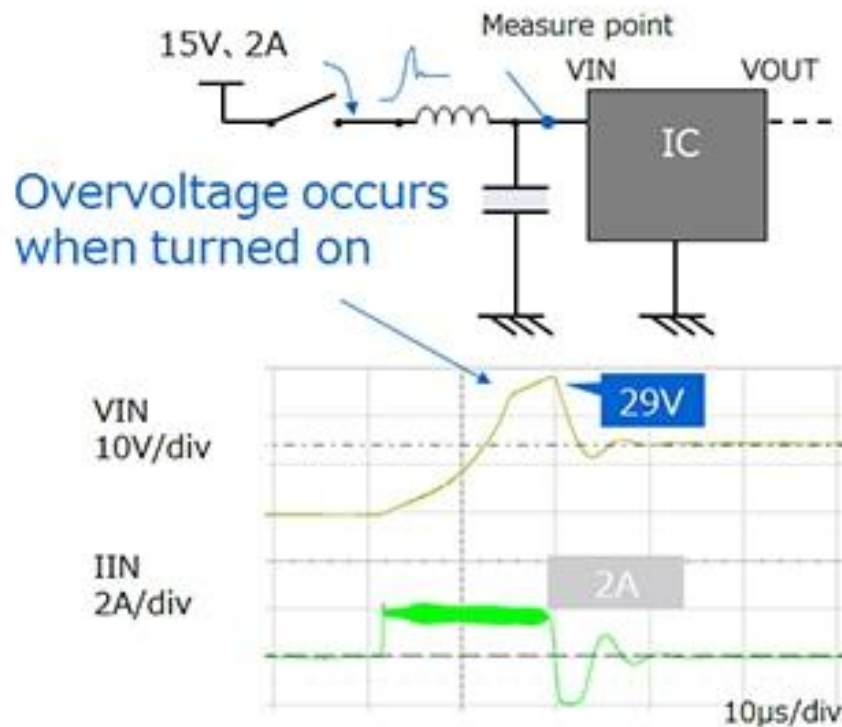
Zener diodes application examples:



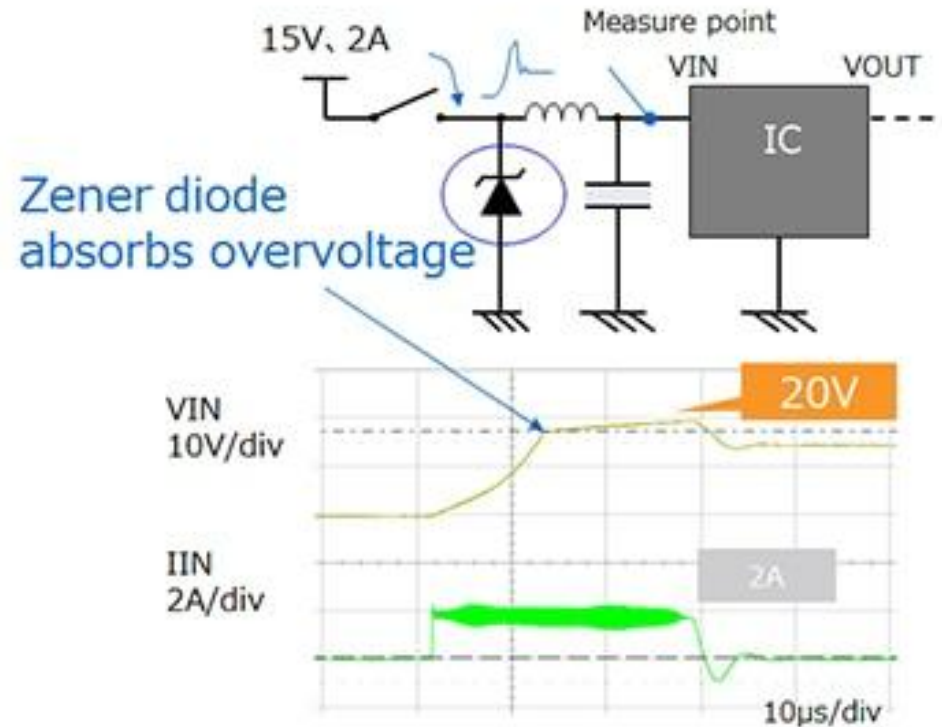
Single Zener Diode Clipping



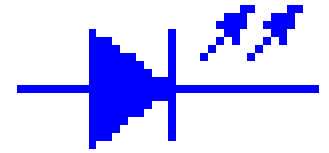
Without Zener Diode



With Zener Diode

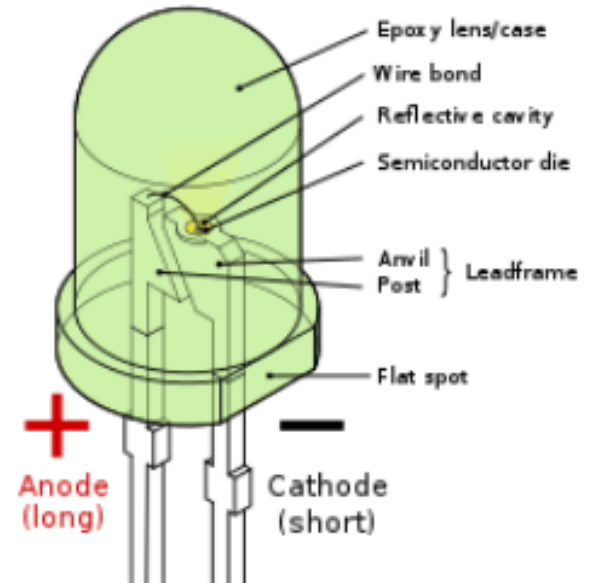
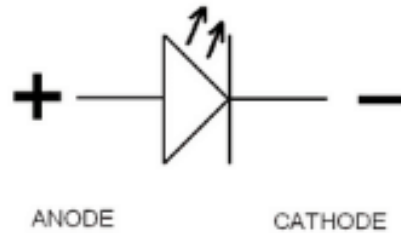
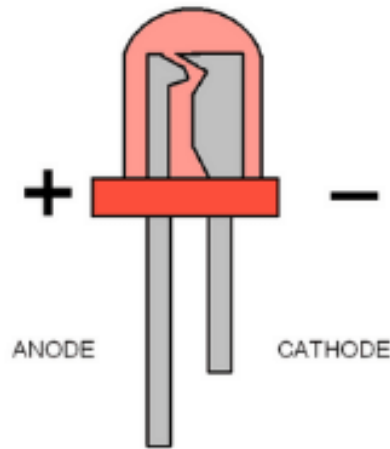


LED Light Emitting Diode



- At a forward bias of junction, electrons from n region get recombined with holes from p region, generating energy as heat and light
- Doping regions with impurities, the emitted light's wavelength may be controlled, obtaining different LED colors
 - GaAsP gallium-arsen-phosphorus – red/yellow light
 - GaP – red/green light
- Direct voltage is higher than for Si diodes (1,2V - 3,2V)
- Direct current is higher also ($\geq 10\text{mA}$)

What is a LED?



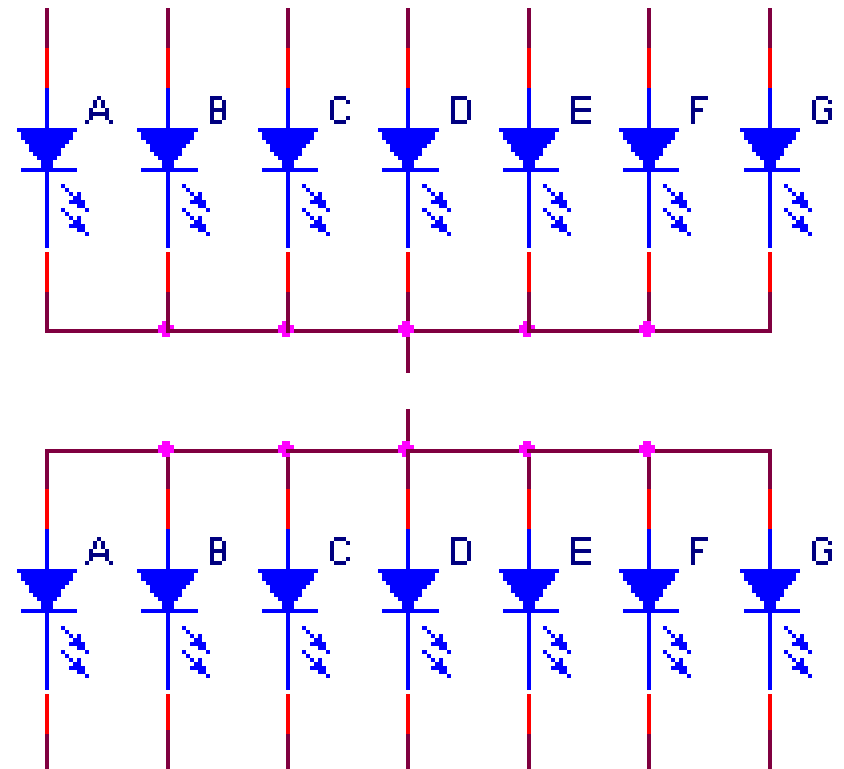
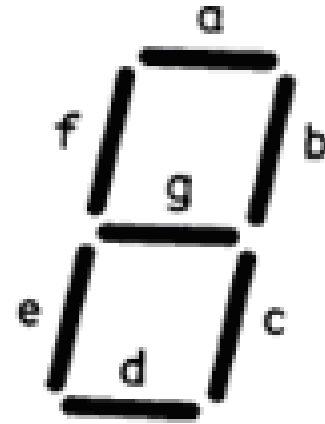
Electrical 4 U

THD LED Diode

SMD Power LED Diode

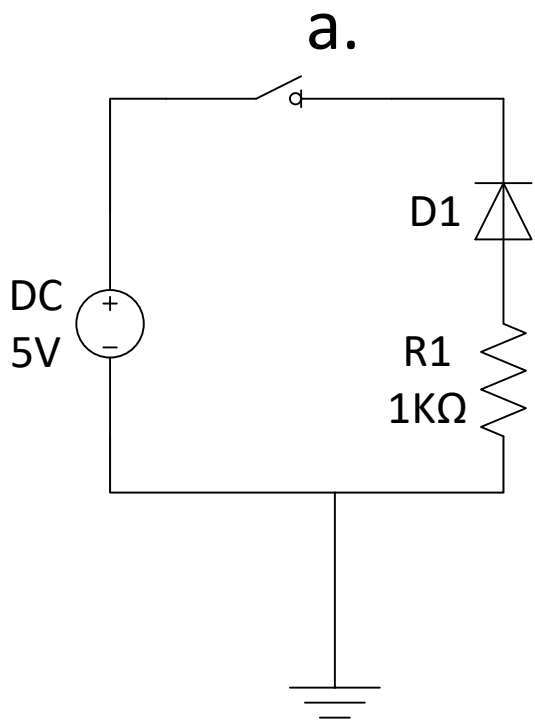


- 7 segments display: 7 LEDs in an array displaying the decimal digits
- Two types:
 - Common cathode – LEDs are driven in anode with an active '1' signal
 - Common anode – signal applied in LEDs cathode, with active low level

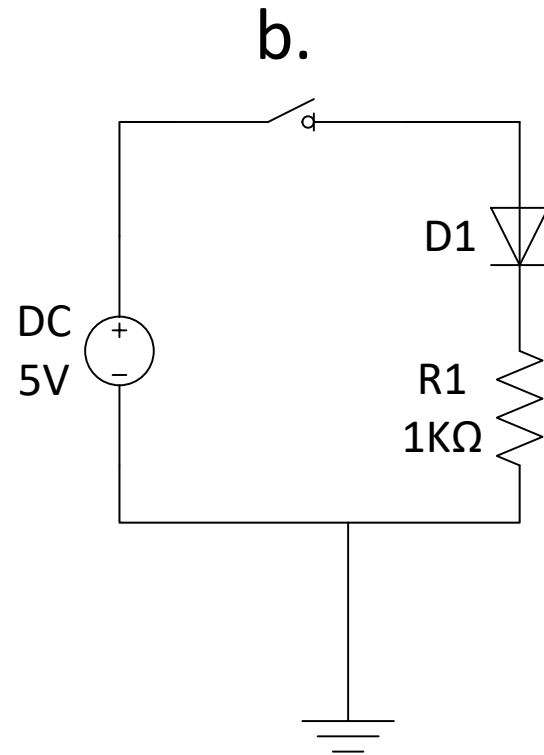


Problems

- For the diagrams in the figure, calculate the value of the current through the diode.

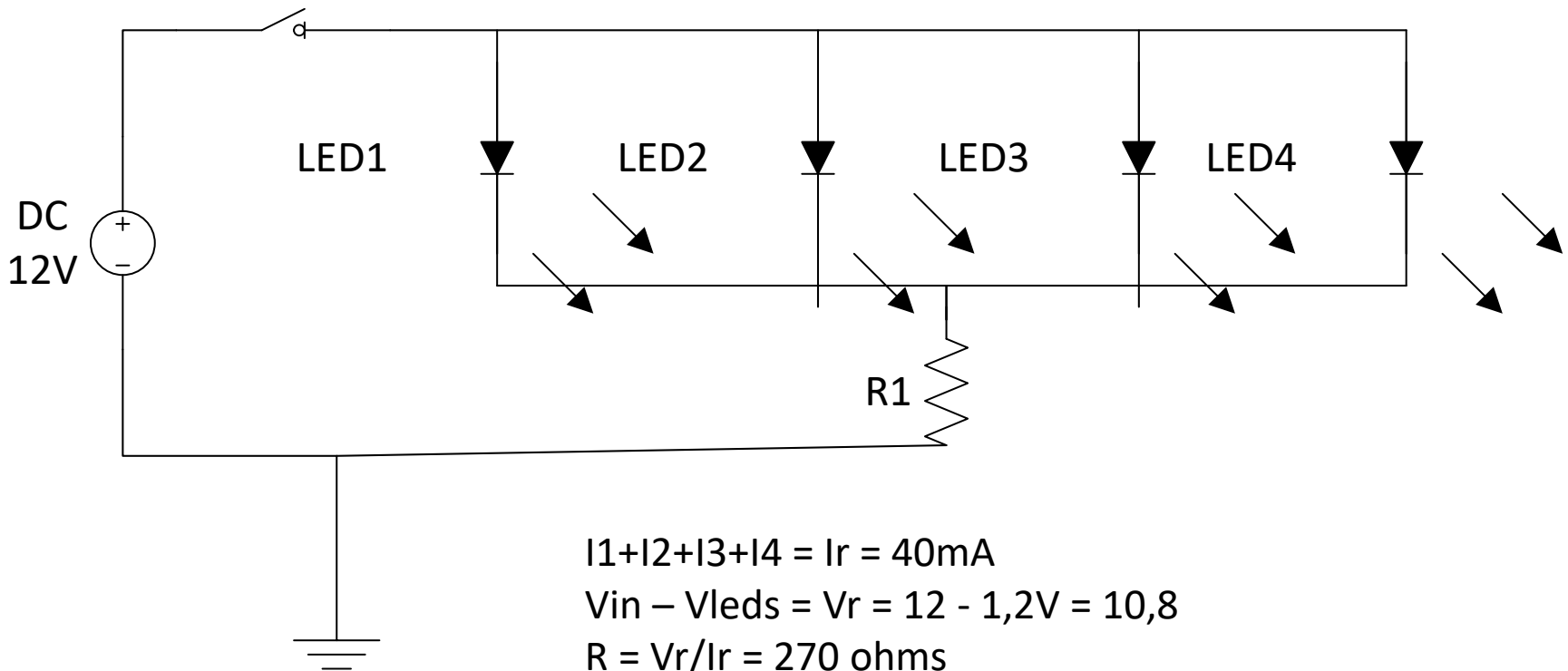


a. D1 reversed biased – open circuit
 $\Rightarrow I = 0$ / negligible



b. D1 forward biased – closed circuit
 V_t for D1 = 0.7V
 $I = V/R = (V_{in} - V_t)/R_1 = (5 - 0.7)/1000 = 4,3 \text{ mA}$

- For the diagram in the figure, compute the value of the resistance so that through each LED the current is 10 mA

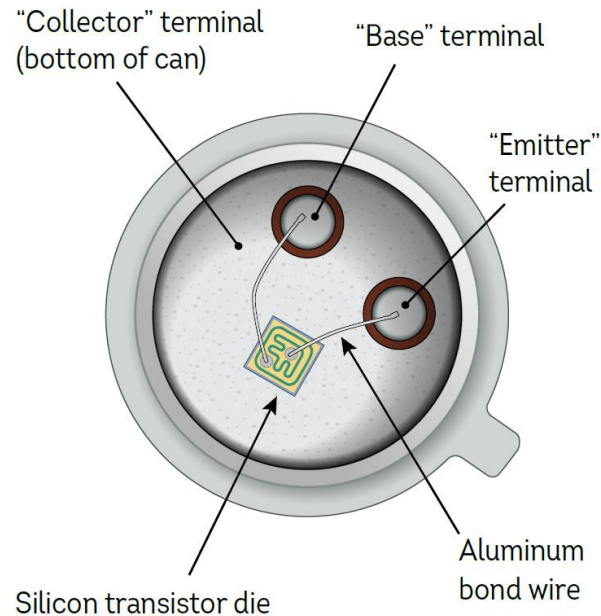


Bipolar junction transistor (BJT)

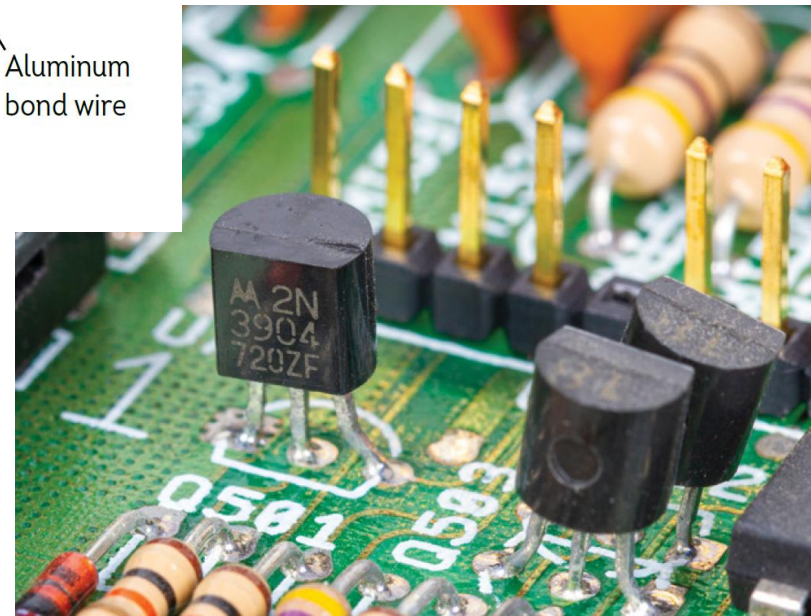
- Semiconductor device
- Commonly used to amplify signals or as logical switches.



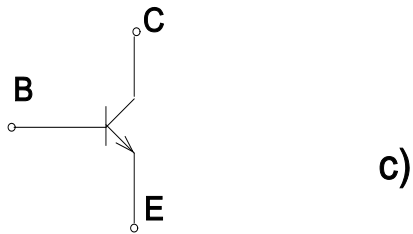
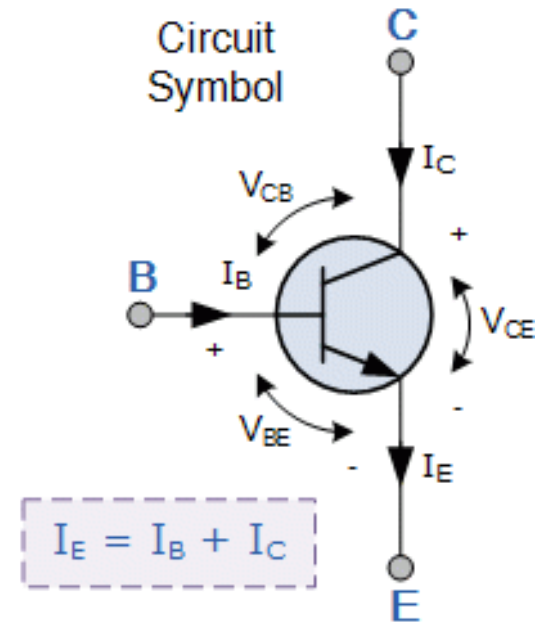
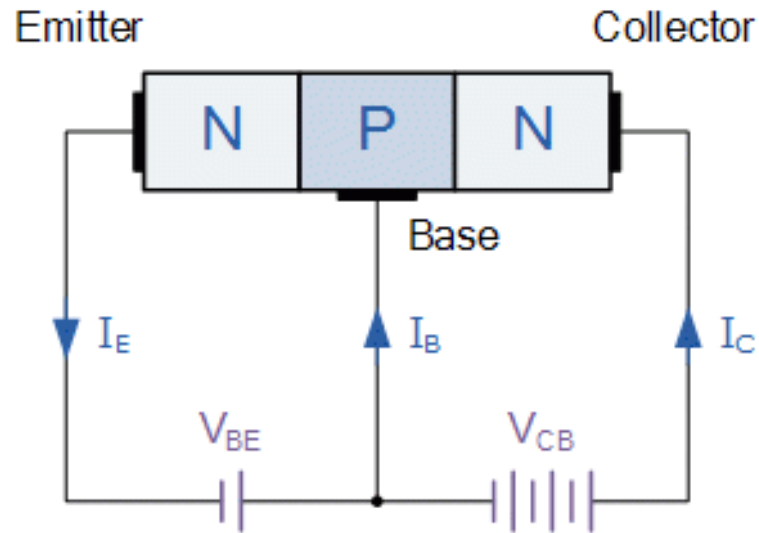
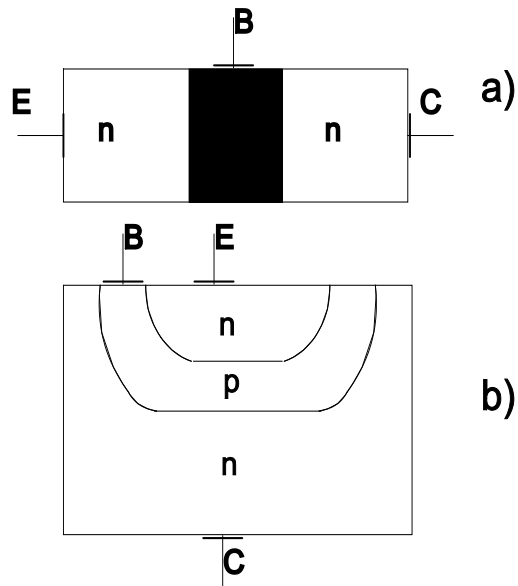
2N2222



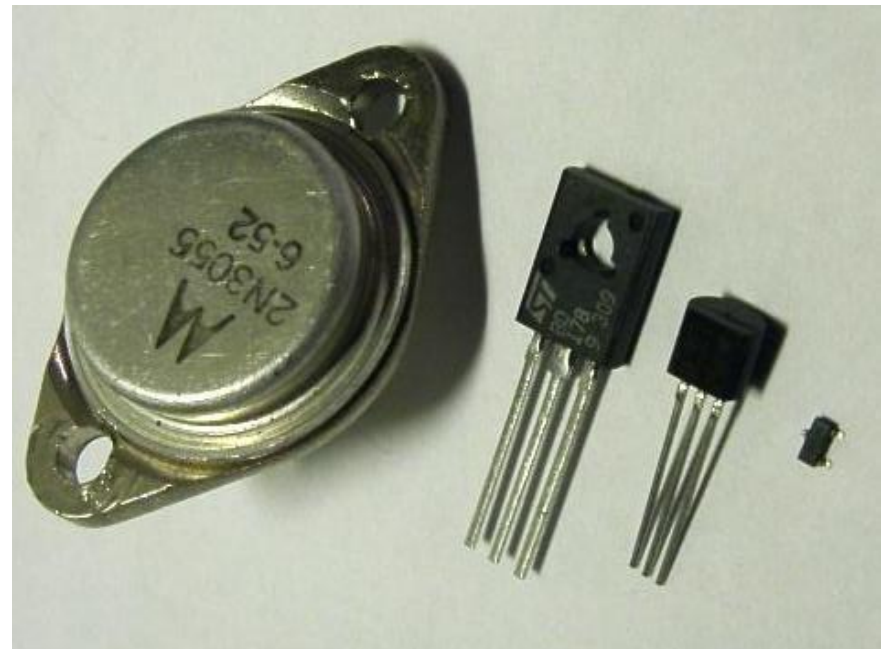
2N3904



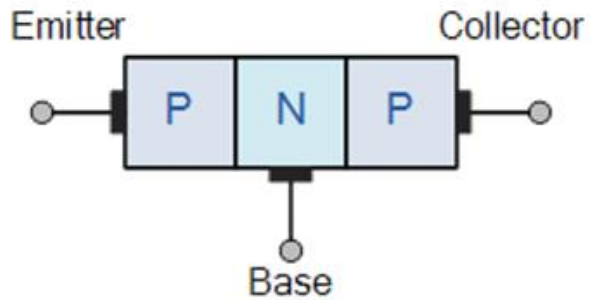
Bipolar junction transistor



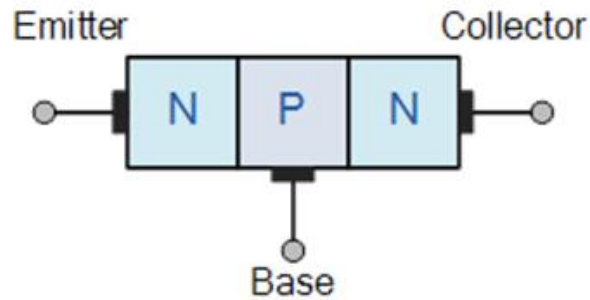
The structure of a bipolar transistor with junctions and some examples



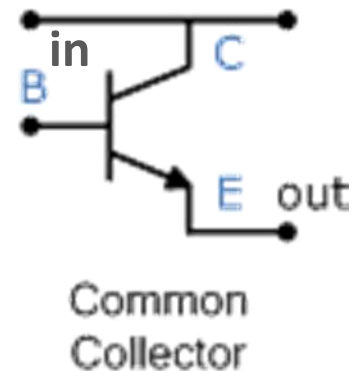
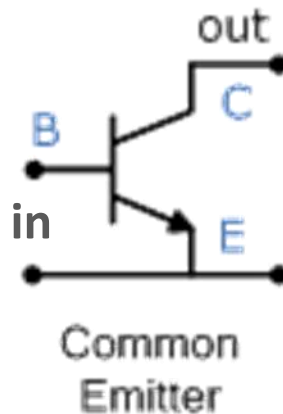
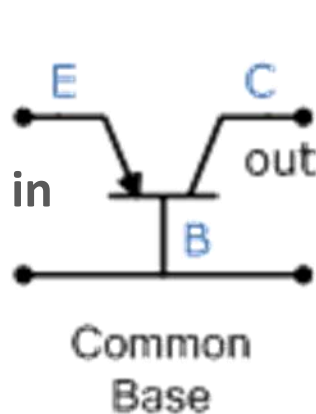
PNP Transistor



NPN Transistor

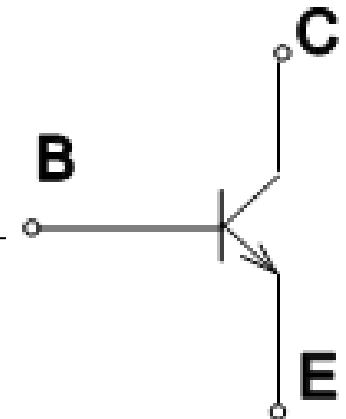


Characteristic	Common Base	Common Emitter	Common Collector
<i>Input Impedance</i>	Low	Medium	High
<i>Output Impedance</i>	Very High	High	Low
<i>Phase Shift</i>	0°	180°	0°
<i>Voltage Gain</i>	High	Medium	Low
<i>Current Gain</i>	Low	Medium	High
<i>Power Gain</i>	Low	Very High	Medium



Functioning regimes of a bipolar transistor

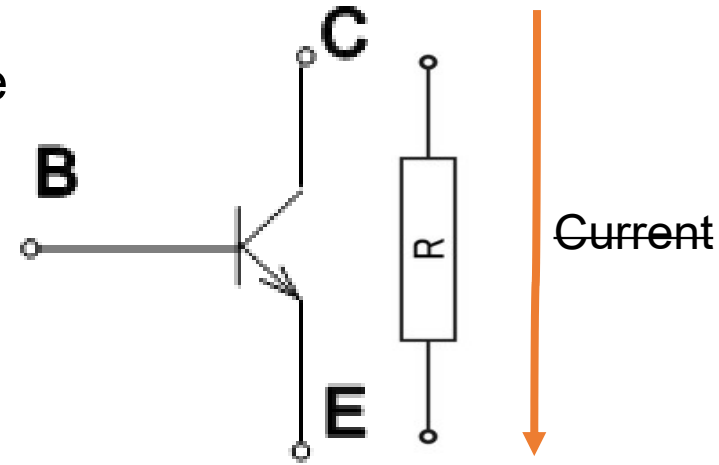
	Base-emitter junction, forward bias ($V_B > V_E$)	Base-emitter junction, reverse bias ($V_B < V_E$)
The base-collector junction, forward bias ($V_B > V_C$)	The saturation region $V_{BE} > V_{CE}$ $I_E = I_C + I_B$ $I_C < \beta \cdot I_B$ $V_{BE} \approx 0,75V$ $V_{CEs} \approx 0,2V$	The reverse active region α_i – very small $(\alpha_i \approx 0,1)$
The base-collector junction, reverse bias ($V_B < V_C$)	The forward active region $I_E = I_C + I_B$ $I_C = \alpha \cdot I_E$ $I_C = I_B \cdot (\alpha / (1 - \alpha)) = \beta \cdot I_B$ $(\beta: 10-1000)$ $V_{BE} \approx 0,75V$ $(V_C > V_B > V_E)$	The cut-off region $V_{BE} \leq 0$ $V_{CE} - V_{BE} > 0$



Bipolar transistor: is a **current operated** device

A. If the BJT is closed ($V_{BE} < 0,6V$)

$\Rightarrow R_{CE}$ has a very high value and V_{CE} can have different values, depending on other components in the circuit (can have large values)



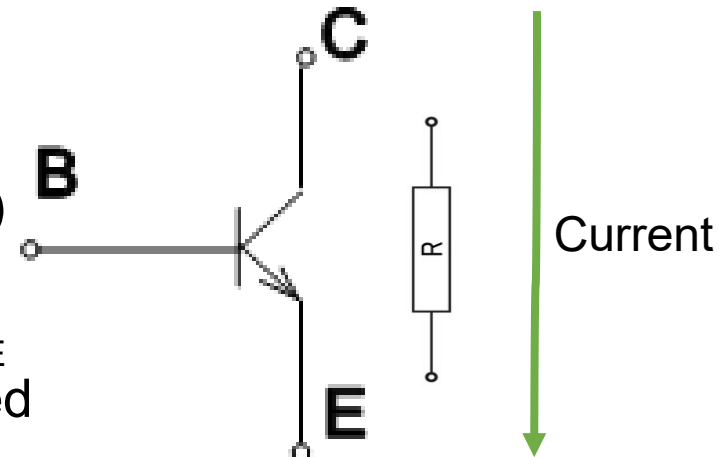
B. If the BJT is opened

B1. $V_{BE} = 0,6V$

$\Rightarrow R_{CE}$ value will decrease (however, still high value) and the current will flow between C and E; V_{CE} can have large values

B2. V_{BE} between 0,6V and 0,75V (V_{BE} will be limited to 0,75V so not to damage the junction)

$\Rightarrow R_{CE}$ value will be small $\Rightarrow$ the current between C and E will increase its value; V_{CE} will drop; when the BJT is completely opened (at saturation) V_{CE} will drop to its minimum value: $V_{CEs} \approx 0,2V$



The most important relations for **forward active region** are:

$$(V_C > V_B > V_E)$$

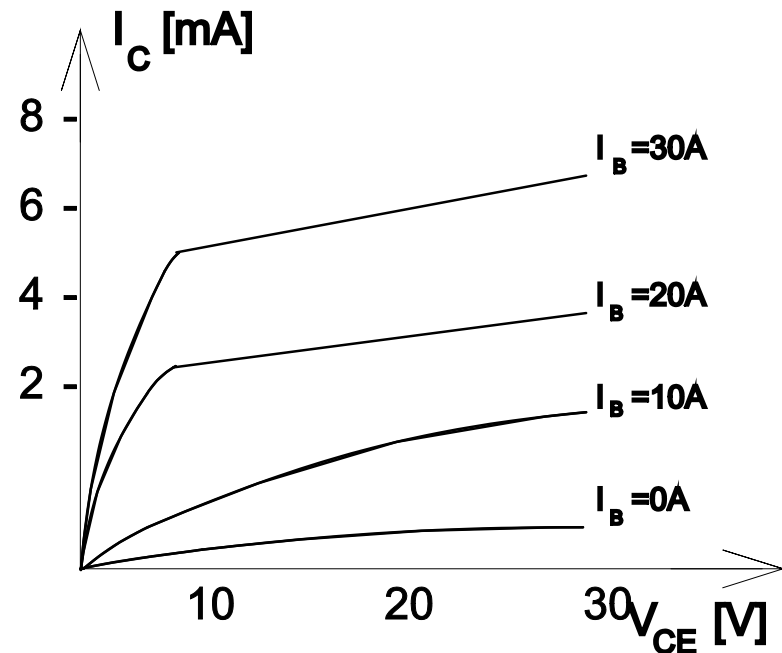
$$I_C = \alpha \cdot I_E \quad (\alpha = I_C / I_E)$$

-- α is the gain in current of the transistor with the common base

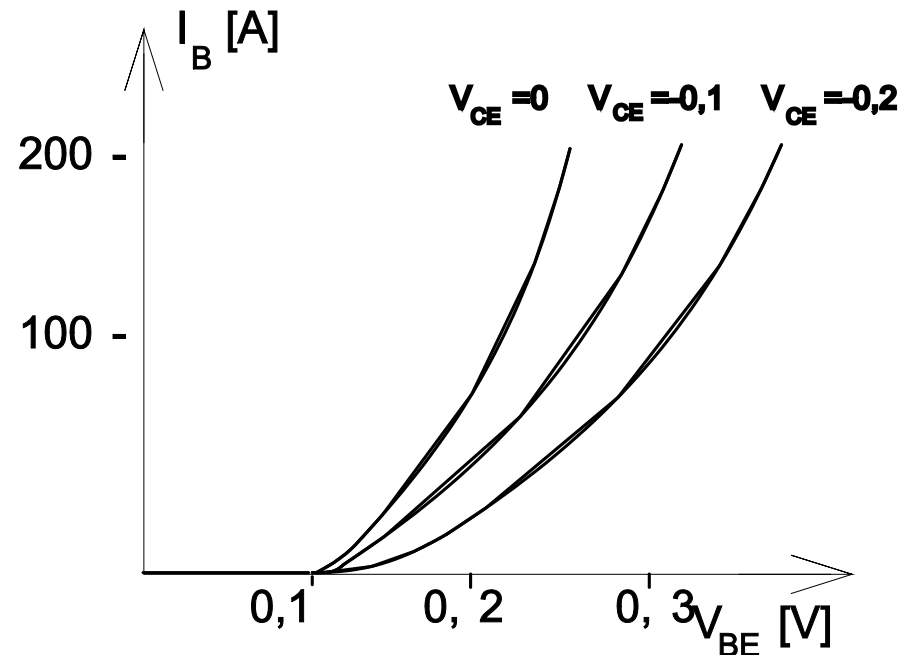
Applying a current I_B into the base, a collector current will result:

$$I_C = I_B \cdot (\alpha / (1 - \alpha)) = \beta \cdot I_B. \quad (\beta = I_C / I_B)$$

-- β is the current gain for a common emitter configuration, or amplification of the current (values from 10 to 1000)



Output characteristic



Input diagram

Cut-off region when both junctions are reverse biased;

State is defined by relations:

$$V_{BE} \leq 0$$

$$V_{CE} - V_{BE} > 0$$

Saturation region implies that both junctions to be forward biased

The relations for the saturation regime are:

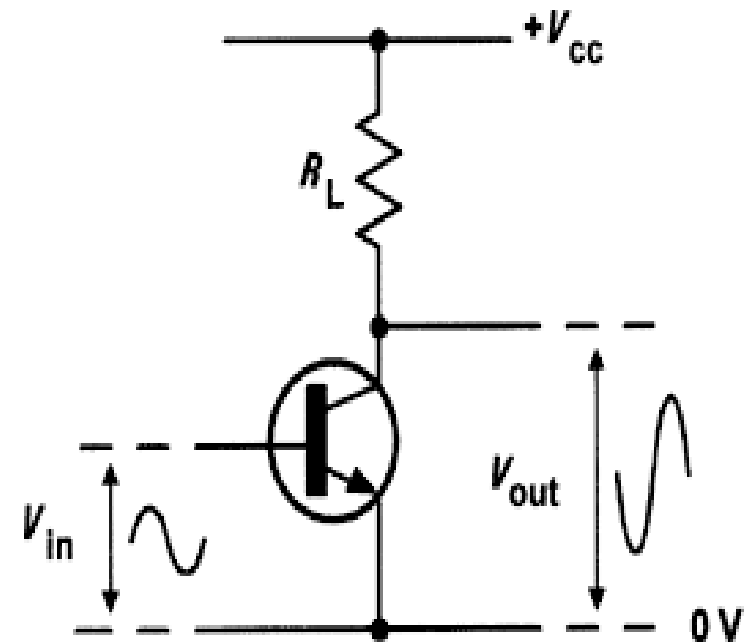
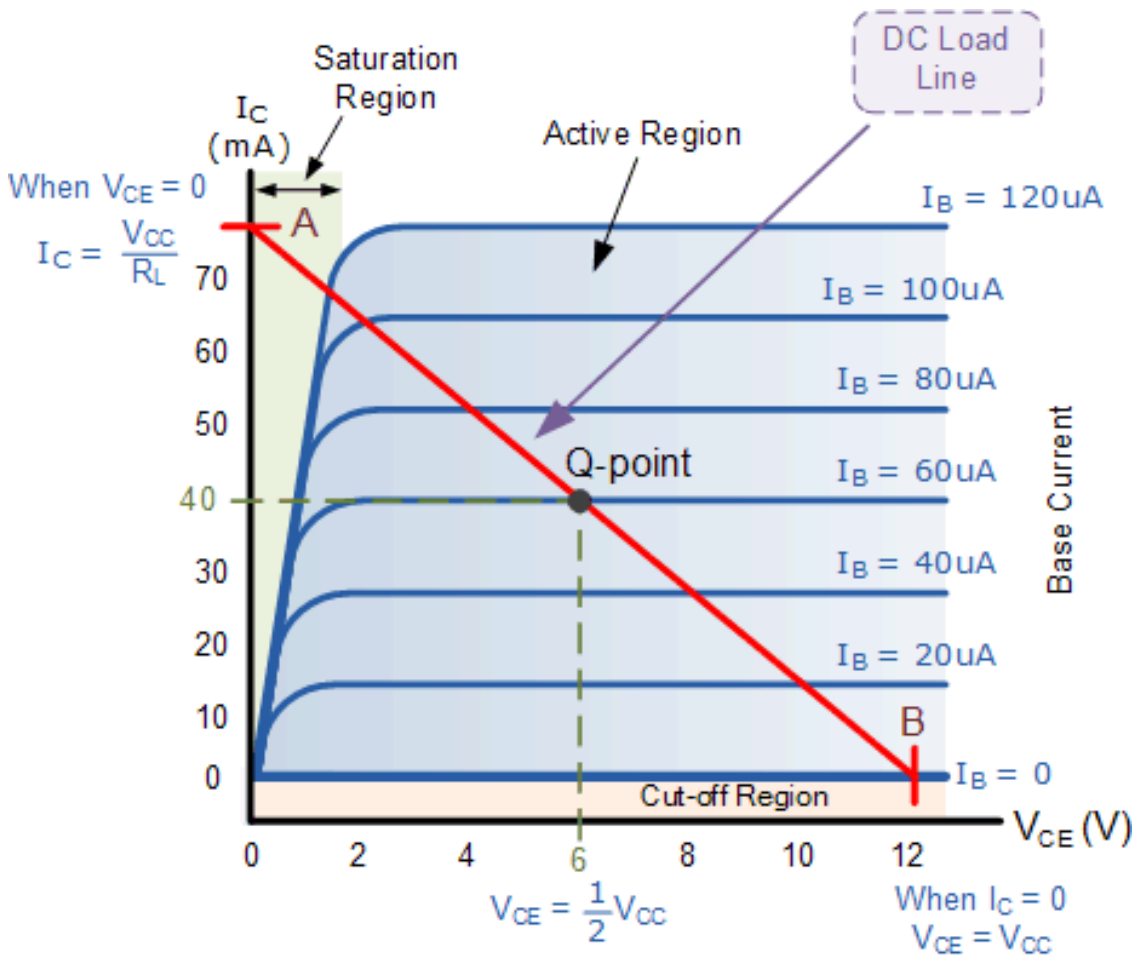
$$V_{BE} > V_{CE}$$

$$I_C < \beta \cdot I_B$$

$$V_{CEs} \approx 0,2V$$

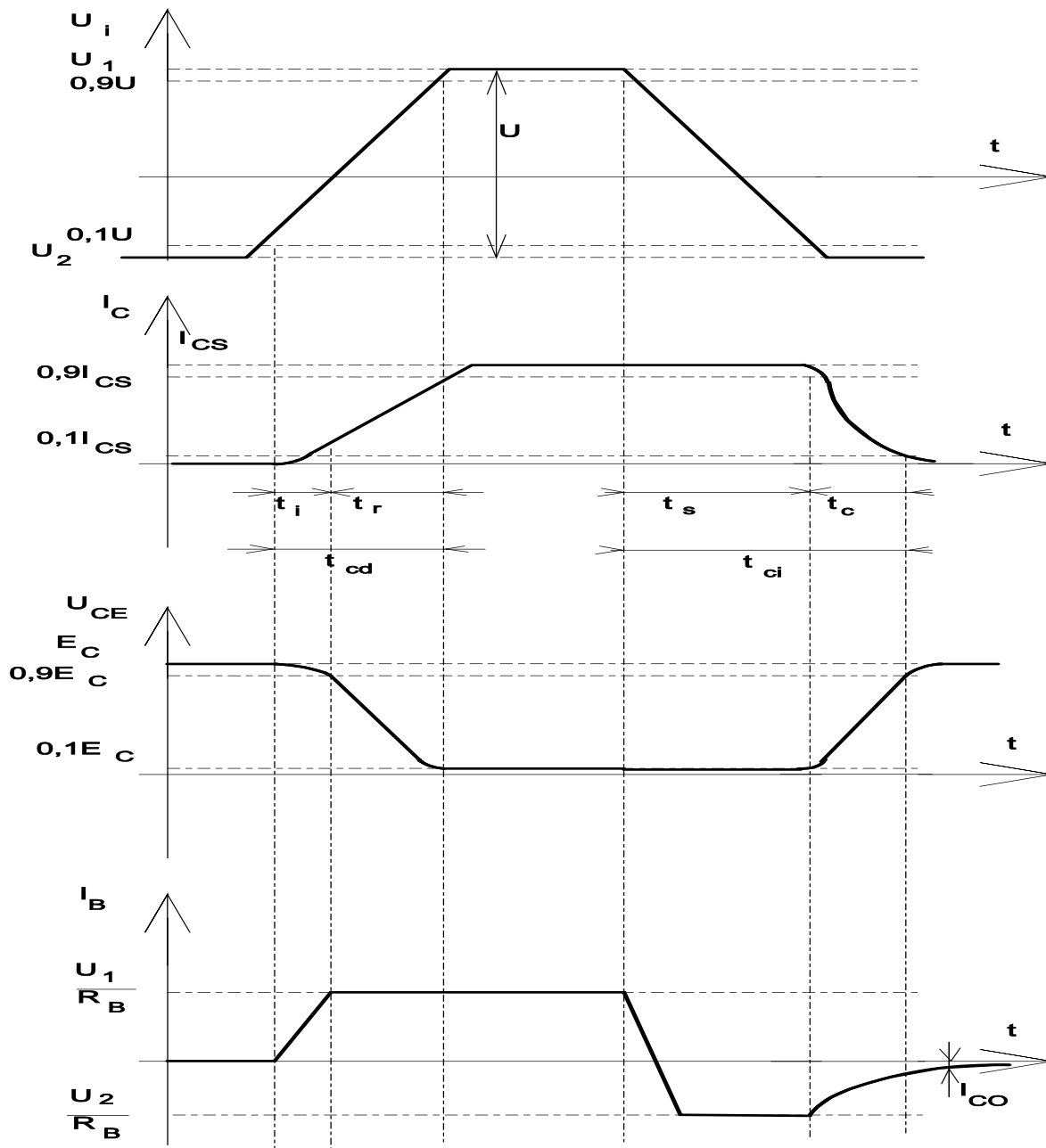
Reverse active region is equivalent with a normal behavior in active region, in which the roles of the emitter and the collector are reversed

- Rarely used because the current amplification has a very small value ($\alpha_i \approx 0,1$)



Operation points of the switching regime

- A is the saturation and B is the cut-off driving point
- $V_{CE} = V_{CC} - R_L I_C \Rightarrow V_{CEmax}$ if $I_C = 0$
- $I_C = (V_{CC} - V_{CE}) / R_L \Rightarrow I_{Cmax}$ if $V_{CE} = 0$



Forward and reverse switching dynamic parameters

Dynamic switching parameters of the bipolar transistor

Forward switching time t_{cd}

- is defined as the necessary time for a transistor to switch from cut-off to conduction (this includes saturation)

Two components:

- t_i , - delay time
- t_r , - raising time

The **delay time** t_i is composed by three components:

- Necessary time to **load the capacity of the base-emitter junction**, from the initial value (U_2), to the corresponding value for the start of the forward bias region
- Necessary time for the **minority carriers to pass through the base**.
- Necessary time for the **value of the collector current to increase** from I_{C0} (or 0), to $0,1 \cdot I_{Cs}$

The values of these times, which are components of the delay time, are small, negligible, the rising time being the most important

Raising time, denoted by t_r is defined by the time interval in which the collector current increases in value from $0,1 \cdot I_{Cs}$ to the value $0,9 \cdot I_{Cs}$

Time is driven by the applied direct base current, and usual this is the direct base current for approaching point S (start of saturation)

$$I_{Bds} = \frac{I_{Cs}}{\beta_{N0}}$$

To foster unblocking, use of: over-driving factor

$$N_d = \frac{\beta_{N0} I_{Bd}}{I_{Cs}}$$

New value:

$$t_r = \frac{\beta_{N0}}{2\pi f_{\alpha 0}} \ln \frac{1 - \frac{1}{10} N_d}{1 - \frac{9}{10 N_d}} \approx 3 \frac{\beta_{N0}}{2\pi f_{\alpha 0}}$$

Reverse switching time

Switch of transistor from a conduction state (forward active/saturation) to a blocking (off) state

Time called t_{ci} it has two components:

storage time t_s ,

falling down time t_c

Storage time t_s has components:

t_{s1} represents the time necessary to **eliminate the excess of charges from the base**, such that the transistor would be in the forward active region

t_{s2} represents the **time needed for the collector current to decrease** from I_{Cs} to $0,9 \cdot I_{Cs}$

$$t_{s1} = \tau_s \ln \frac{I_{Bd} - I_{Bi}}{I_{Bs} - I_{Bi}} \quad t_{s2} = \tau_r \ln \frac{I + \frac{N_b}{I}}{I + \frac{0,9 N_b}{I}}$$

τ_s is the storage time constant

I_{Bd} is the forward base current

I_{Bi} is the reverse base current

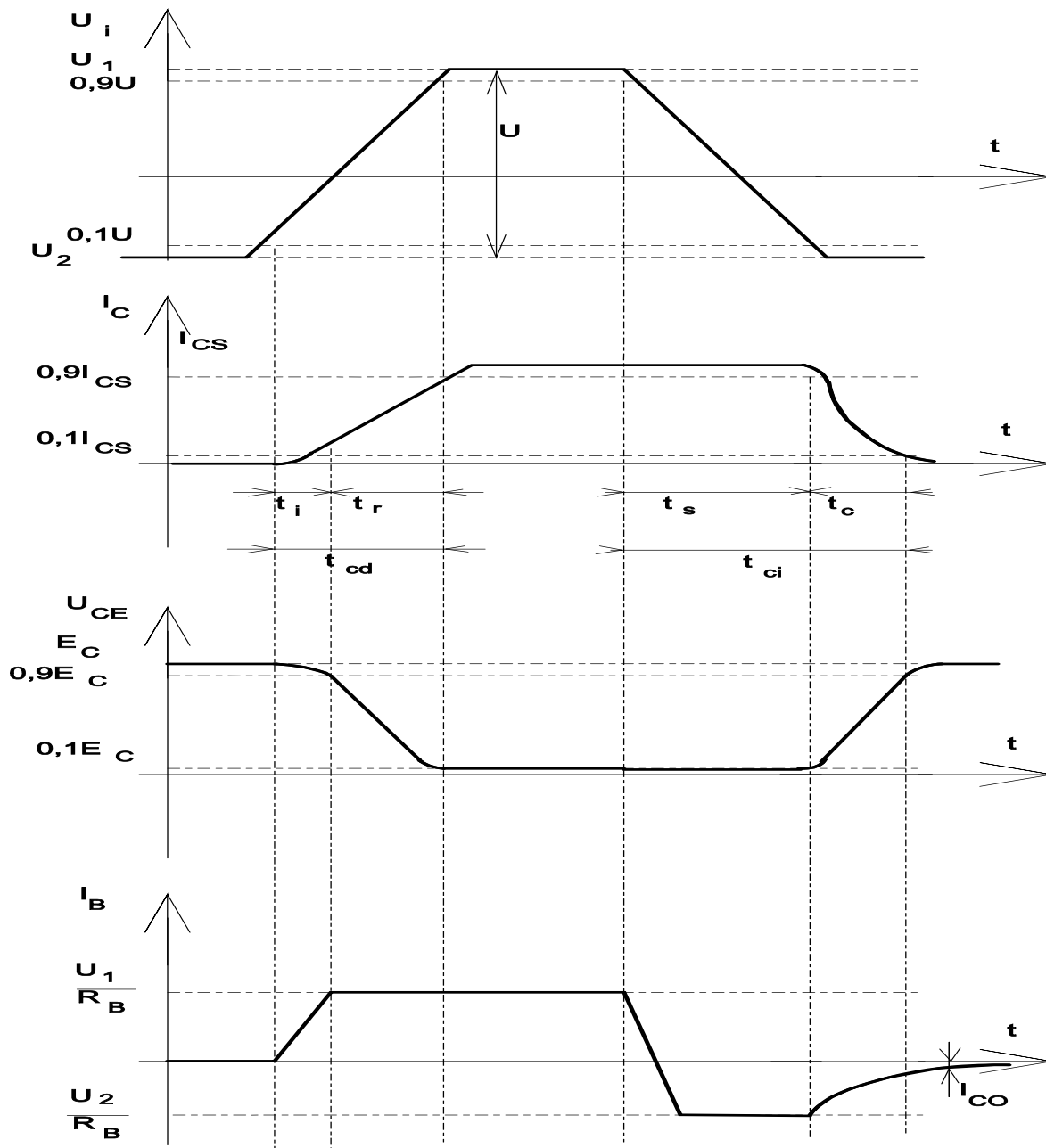
I_{Bs} is the base current at the boundary of the saturation and the forward active regions

t_c , falling down time (necessary for the collector current to decrease from $0.9I_{Cs}$ to $0.1 I_{Cs}$):

$$t_c = \tau_r \ln \frac{1 + \frac{9}{10 N_b}}{1 + \frac{1}{10 N_b}}$$

N_b represents the blocking over-driving coefficient

$$N_b = \frac{\beta_{N0} I_{Bi}}{I_{C0}}$$



Forward and reverse switching dynamic parameters

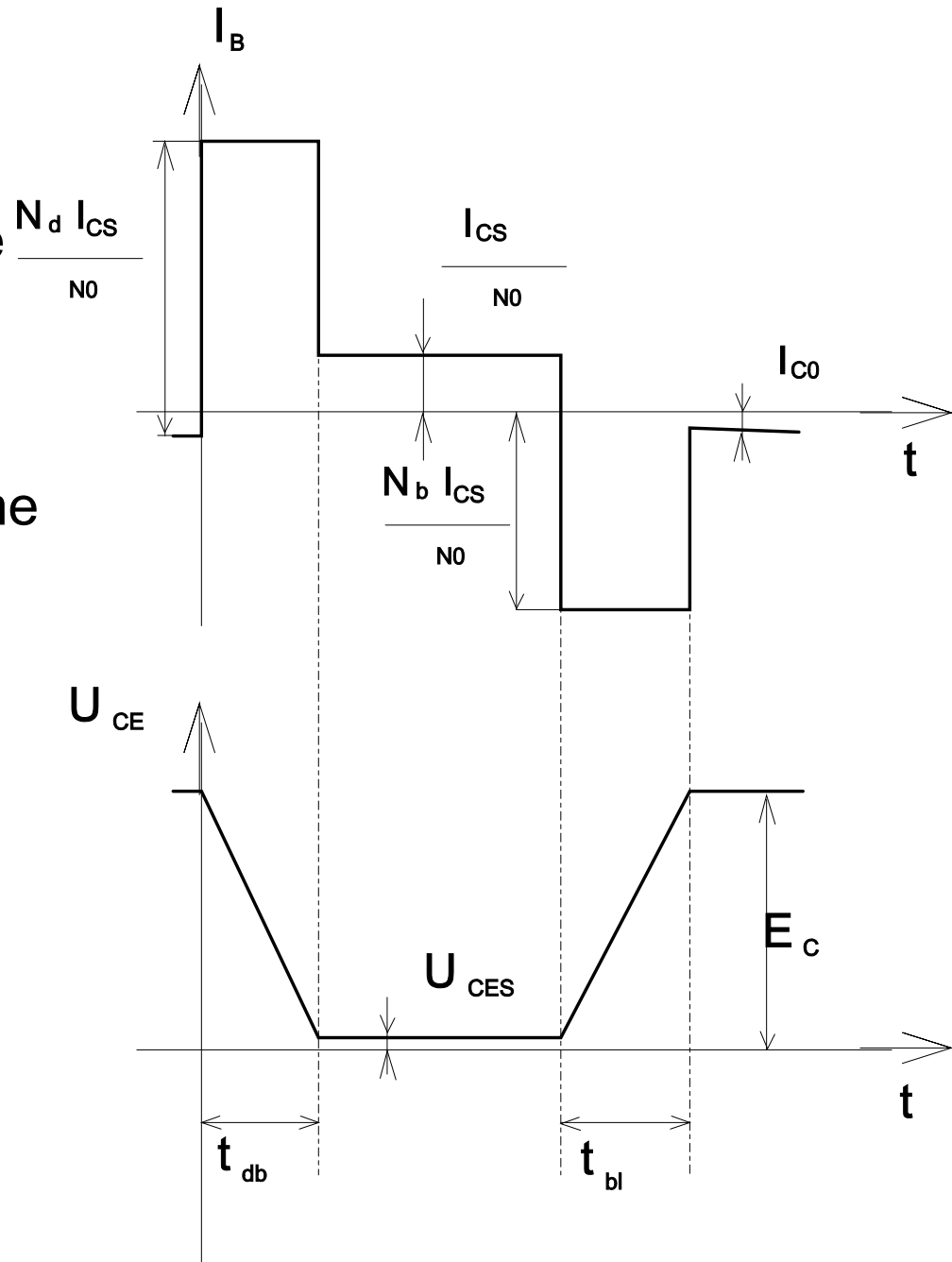
Speed-up methods

Following methods to reduce the switching time for a transistor:

- over-drive at unblocking, to decrease the direct switching time

- over-drive at blocking, to decrease the reverse switching time

- avoid the saturation stage by canceling the storage time



Speed-up capacitances

Apply of a pulse voltage on circuits input
 U_1 high level, U_2 low

For a positive voltage swing:

Time constant for charging up the capacitor is:

$$\tau_{inc} = C \cdot (R_g + R_{in})$$

base current now exponentially decreases, with the same time constant, towards a constant value which does not allow the saturation to take place:

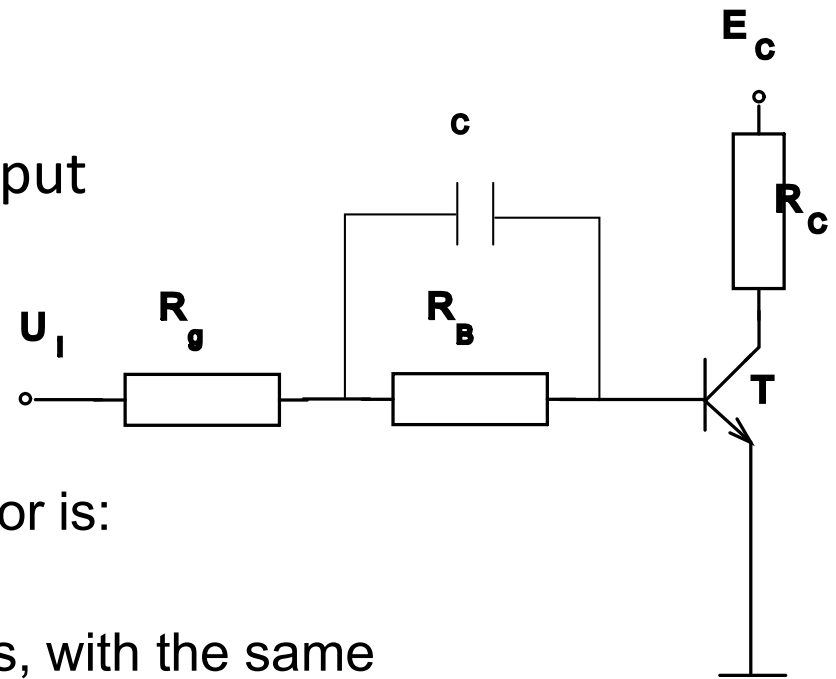
$$I_{Bd} = U_1 / (R_g + R_B + R_{in})$$

For a negative voltage drop:

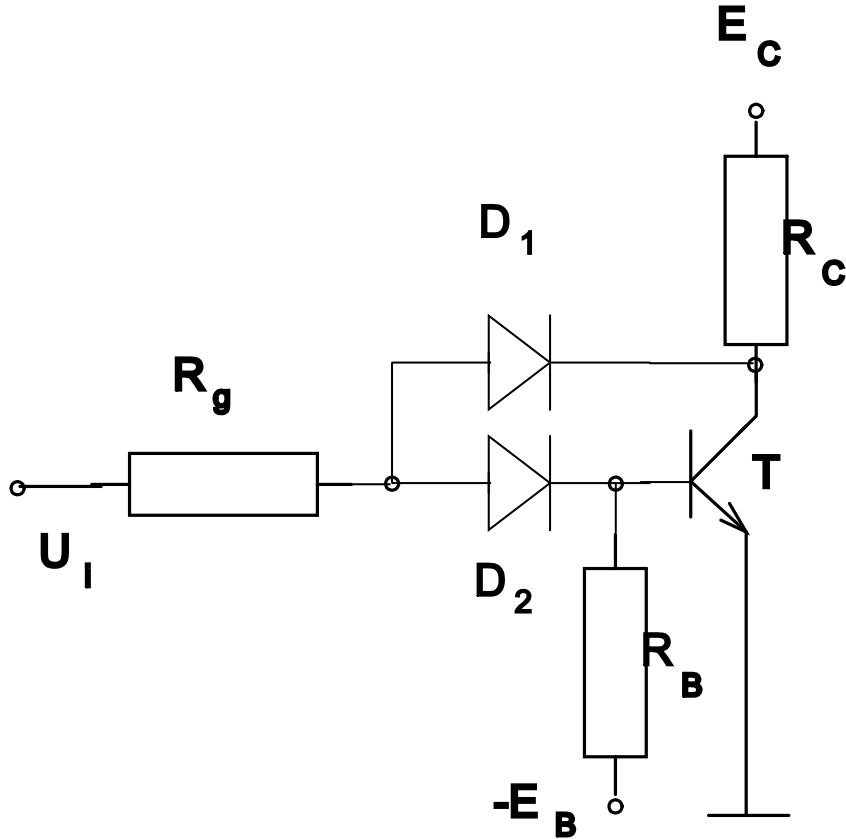
Base current falls to I_{Bi} whose value is determined by U_2 and R_{in}' – the high resistance of the blocked transistor. Then, once the transistor is blocked and the capacitor is unloaded, the reverse current exponentially decreases

towards a constant value: $I_{Bi} = I_{C0}$

Capacitor time constant now: $\tau_{disc} \approx C \cdot R_B$



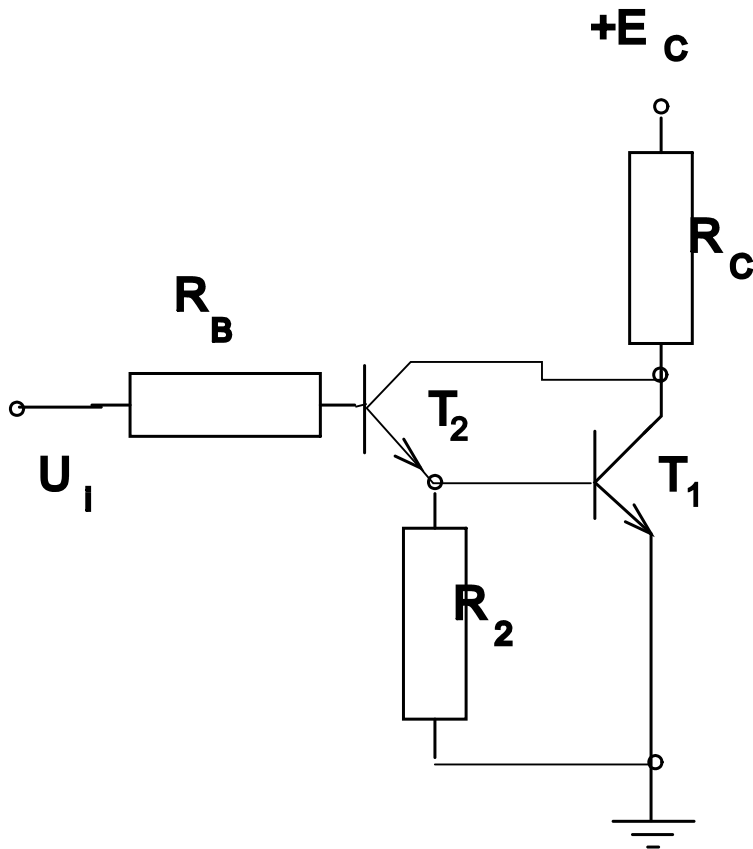
Using a negative, non-linear voltage reaction to avoid saturation



Avoid the saturation of the T transistor by using the diodes D_1 și D_2

- The diodes have different junction potentials

Using a Darlington circuit



T₂ – the control transistor and T₁ – the output transistor

- T₁ never gets saturated as long as the collector potential is always greater than the base potential:

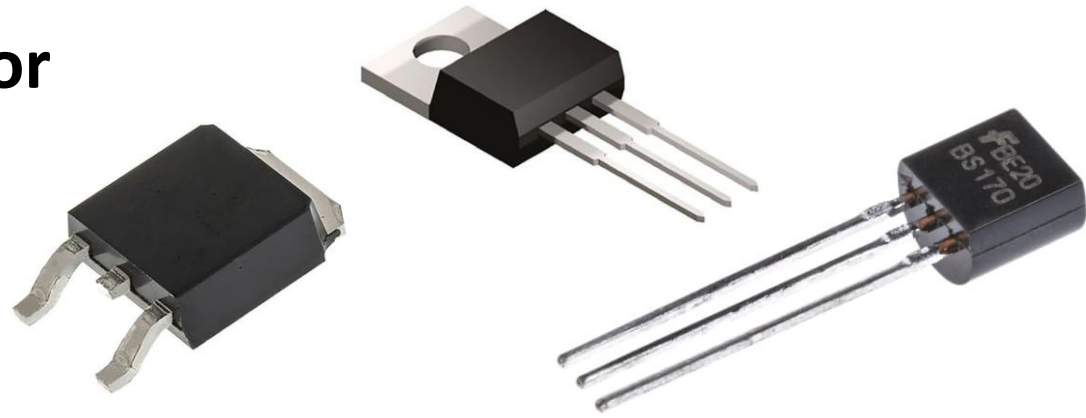
$$U_{C1} = U_{CE2} + U_{B1}$$

Therefore, $V_{C1} > V_{B1}$, and the base-collector junction is reversed biased

The Field Effect Transistor

Classification

- gate junction transistor
- insulated gate transistor
- thin layers transistor



Study of IGFET (also known as MOSFET (*metal-oxide* FET)) or insulated gate transistors

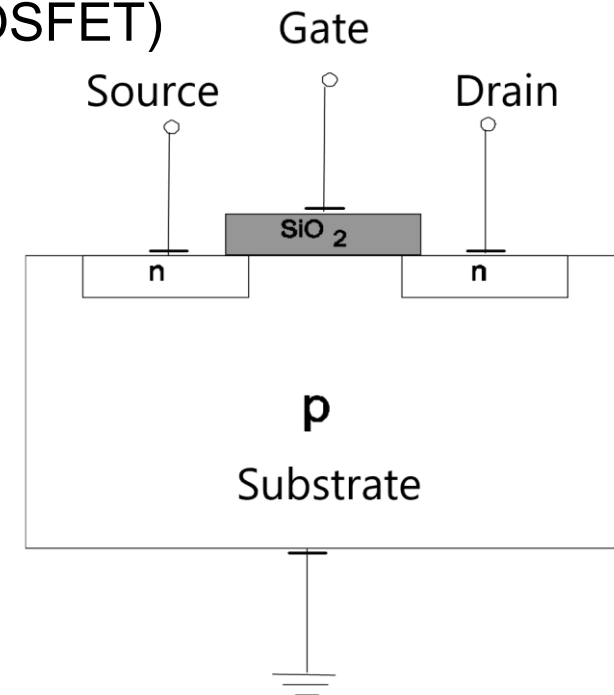
Grouped by the manufacturing technology in:

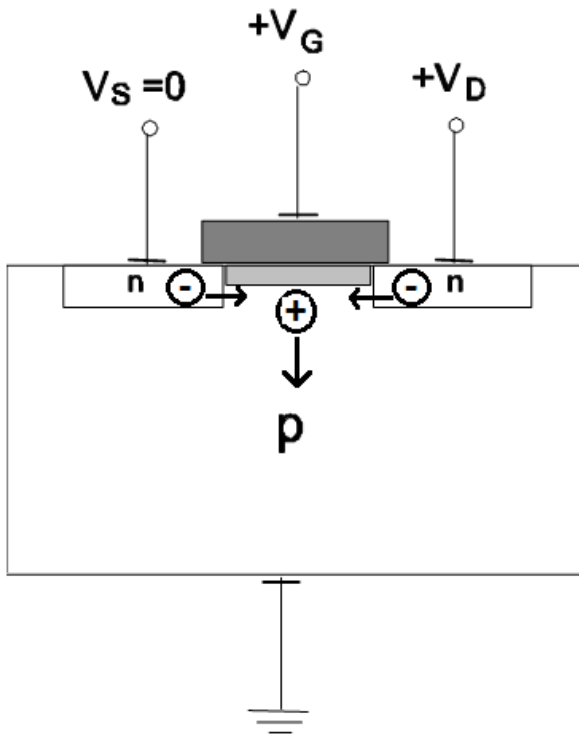
- MOS transistors with induced channel (enhancement MOSFET)
- MOS transistors with initial channel (depletion MOSFET)

The physical structure of a MOS transistor is presented
Shown a cross section of a MOS transistor with
n induced channel, also called NMOS

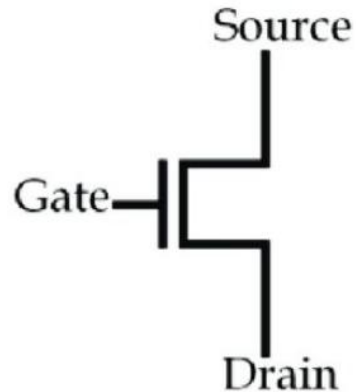
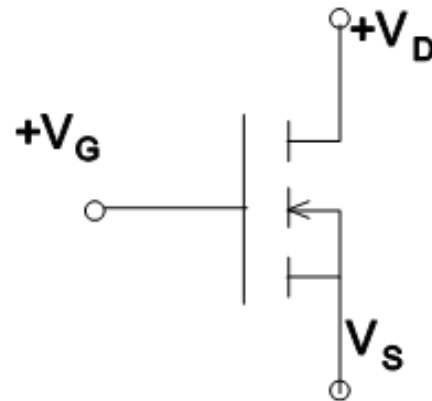
Components:

- basis substrate
- source
- drain
- gate (grid)

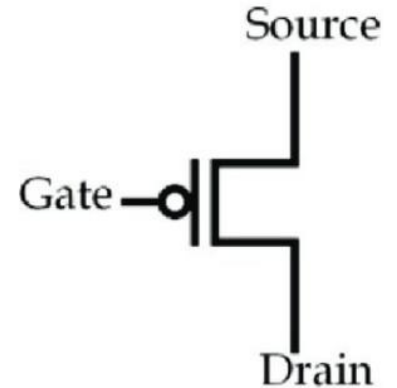
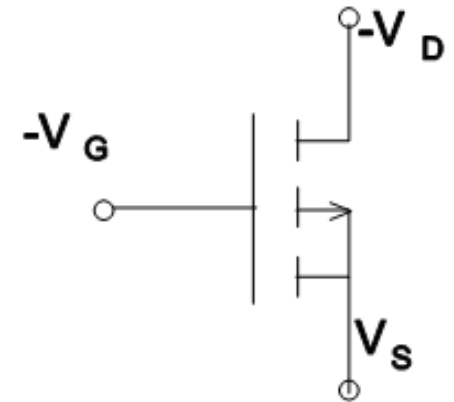




NMOS Electrodes



NMOS



PMOS

FET is a **voltage operated** device: control the electrical current flowing between the Source and Drain contacts, with voltage being applied via the Gate.

NMOS Operation presented next slide

If a positive voltage, relative to the drain and the source regions, is applied on the gate electrode (on the grid) then the electric charges of **p** type from the basis substrate will be rejected and the electrons from drain and source regions will be attracted to the surface of the silicon substrate laying under the gate.

Assume that V_{DS} , the voltage between the drain and the source, is much lower than V_{GS} , the voltage between the gate and the source. Therefore, the potential voltage from this region, laying under the silicon dioxide insulator, is relatively constant. By increasing the grid potential, the intensity of the existing electric field is also increased, as the number of electrons from the region multiplies. We may assume that there exists a channel between the drain and the source whose depth increases proportionally with the voltage applied on the gate. For a **threshold voltage** of V_{GS} , denoted by V_T , the concentration of the electrons will exceed the concentration of the holes. Consequently the region will reverse its type: from a **p** type region it will become an **n** type region. It follows that on the surface of the **p** type semiconductor an **n** type channel has been created. The conductance of the drain-source region and the drain current will increase proportionally with V_{GS} .

If the voltage between the drain and the source decreases, the voltage between the gate and the drain will also decrease. The channel will present an irregular distribution of the electric charges. This denotes a non-linear dependence of I_{DS} with respect to V_{DS}

Advantages of using MOS transistors:

more easily manufactured than bipolar transistors

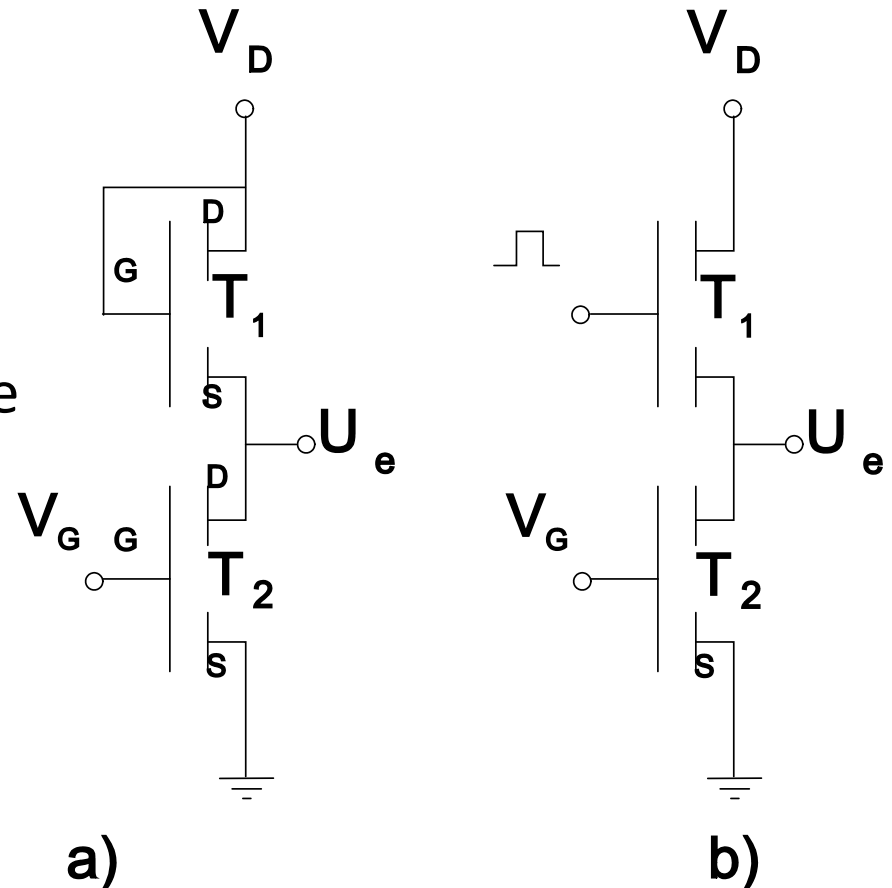
higher density of integration

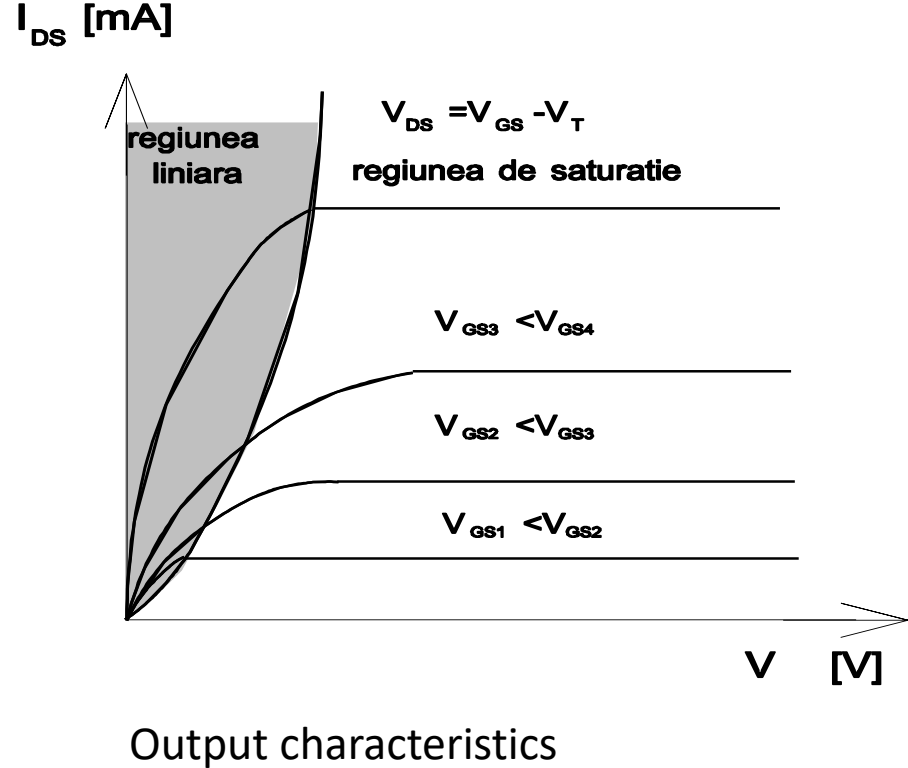
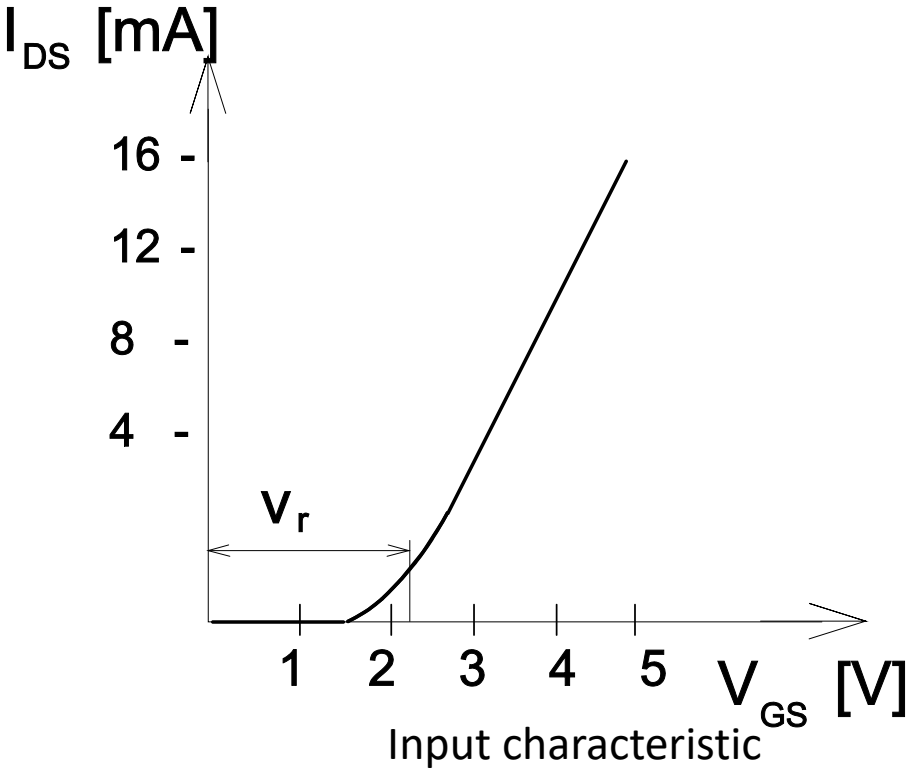
high input impedance (10^{14} - $10^{16}\Omega$), a low driving current

use in MOS structure of an active resistance,

Some disadvantages implied by caution to storage and transport

Ways to implement active resistance





From the output characteristic, three regions can be deduced and analyzed:
the cut-off stage: the output current, the drain-source current I_{DS} is almost null and the input voltage, V_{GS} , is lower than the threshold voltage: $V_{GS} < |V_T|$

the linear (triode) stage: the region situated at the left of the current characteristic, when $V_{DS} = V_{GS} - V_T$; the drain current I_{DS} quickly increases as a function of the drain-source potential V_{DS} :

$$I_{DS} = K((V_{GS} - |V_T|)V_{DS} - \frac{V_{DS}^2}{2})$$

Also: $0 \leq V_{DS} \leq V_{GS} - |V_T|$

-the saturation stage: the region situated at the right of the current characteristic, when $V_{GS} - |V_T| = V_{DS}$

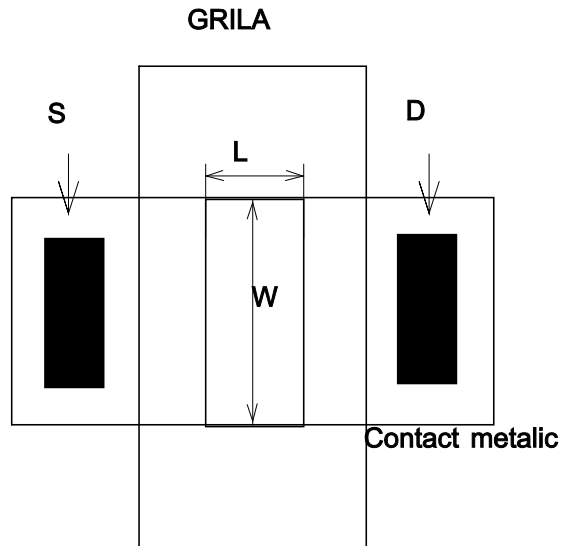
The following relations take place:

$$0 \leq V_{GS} - |V_T| \leq V_{DS}$$

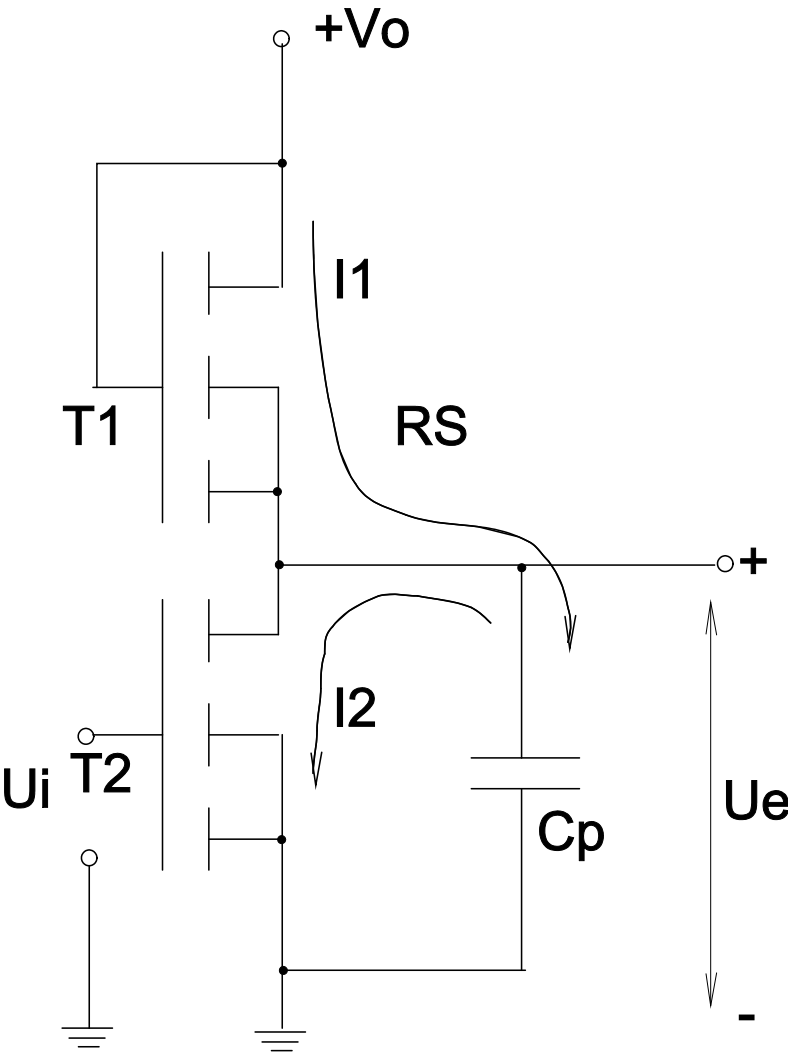
$$I_{DS} = \frac{K}{2} (V_{GS} - V_T)^2$$

K, conduction factor $\approx \beta \cdot (W/L)$, where:

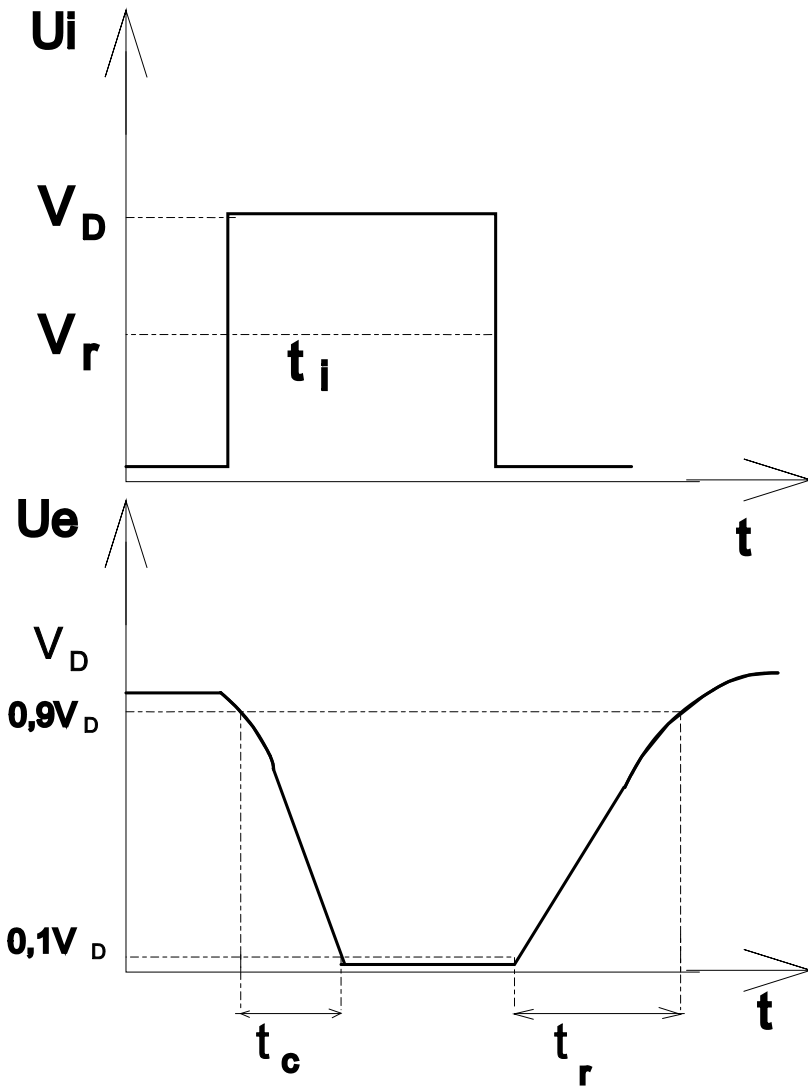
- β is the internal conduction factor; it has an approx value of $10 \mu A/V^2$
- W is the channel width; it takes values in the range of $10-200 \mu m$
- L is the channel length; it takes values between $1-10 \mu m$



Dynamic Parameters for MOS Transistor



Layout of a MOS Inverter



Switching Times

Assume that a MOS transistor has to drive the gate of one or more MOS transistors. It must guarantee the charging and discharging of the input capacitances for the driven transistors. Denote by C_p the sum of all input capacitances of the driven transistors. We may associate the switching times of the driving transistor with the charging/discharging times for the capacity C_p .

The formula for the time constants of the RC circuits are:

$$T_{\text{inc}} = R_s \cdot C_p$$
$$T_{\text{desc}} = R_T \cdot C_p$$

- R_T is the direct resistance of the conducting transistor
- R_s is the load resistance

We choose that the output voltage for the low logic level being as close as possible to the ground potential, $R_s \gg R_T$.

Thus, the raising and falling times for MOS transistors are:

$$t_r = 2,2 \cdot R_s \cdot C_p$$
$$t_c = 2,2 \cdot R_T \cdot C_p$$

Solved application (problem)

Design an inverting circuit using bipolar transistors and passive components

- Stages:
 - Static regime design, checking the fullfillness of proposed logic functions
 - Dynamic regime design, checking the worst case behaviour
 - Dynamic behaviour, estimating the dynamic parameters

Static regime behaviour

- If $U_i = 0V = '0'$, at output:
 $U_e \approx E_C = '1'$, or transistor T be **off** (blocked)
- If $U_i \approx E_C = '1'$, implies
 $U_e \approx 0V = '0'$, or transistor T being saturated

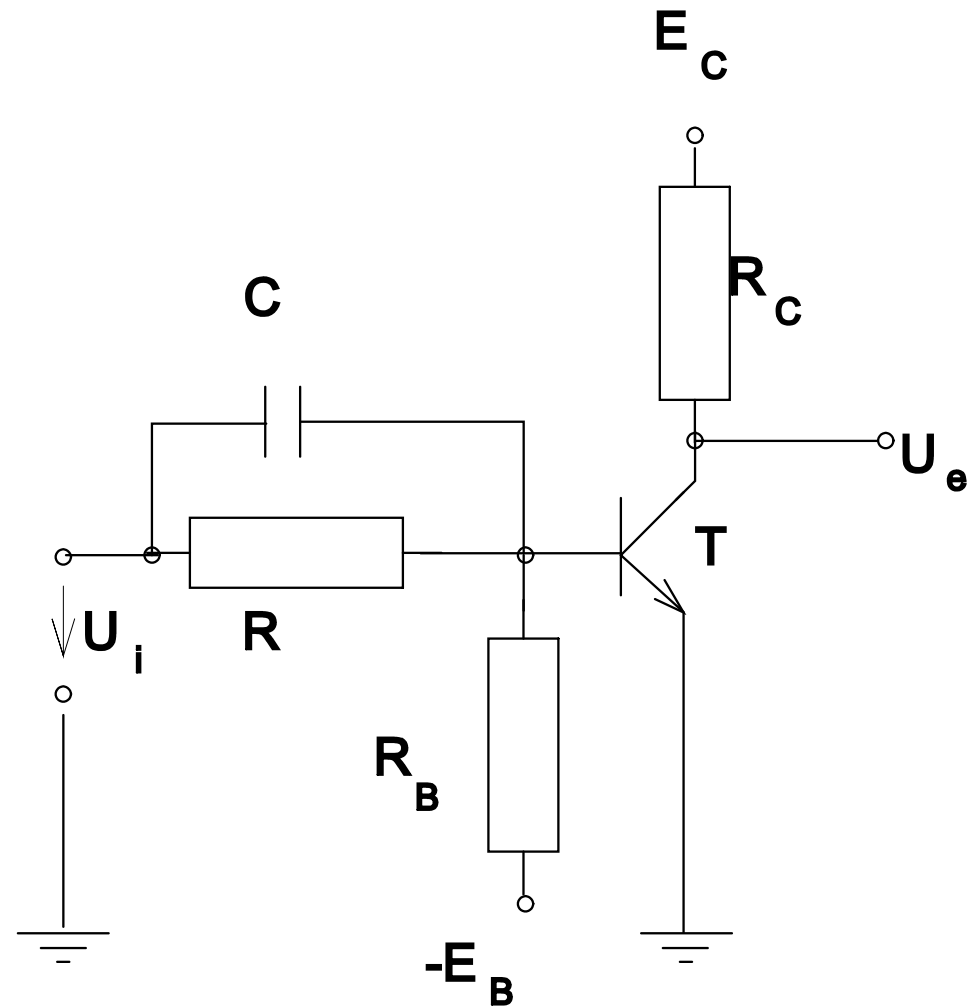
- Off state conditions:

$$U_{BEb} \leq 0V \text{ and } I_B = I_{C0}$$

$$I_R + I_{C0} = I_{RB}$$

$$-\frac{U_{BEb}}{R} + I_{C0} = \frac{U_{BEb} + E_B}{R_B}$$

Because $U_{BEb} \leq 0V$, results
 designed value: $R_B \leq \frac{E_B}{I_{C0}}$



- On state:

$$I_R - I_{RB} = I_B$$

$$I_B \geq I_{Bs} = I_C / \beta_{N0}$$

$$R \leq \frac{E_C - U_{BEs}}{\frac{E_C}{\beta_{N0} R_C} + \frac{U_{BEs} + E_B}{R_B}}$$

- Relations for R and R_B must be valid for $I_{C0} = I_{C0max}$ and $\beta_{N0} = \beta_{N0min}$
- R_C becomes:

$$R_C = \frac{E_C - U_{CEs}}{I_{Cso}}$$

where I_{Cso} – saturation collector current, obtained for a maximum value of β

Worst Case Study

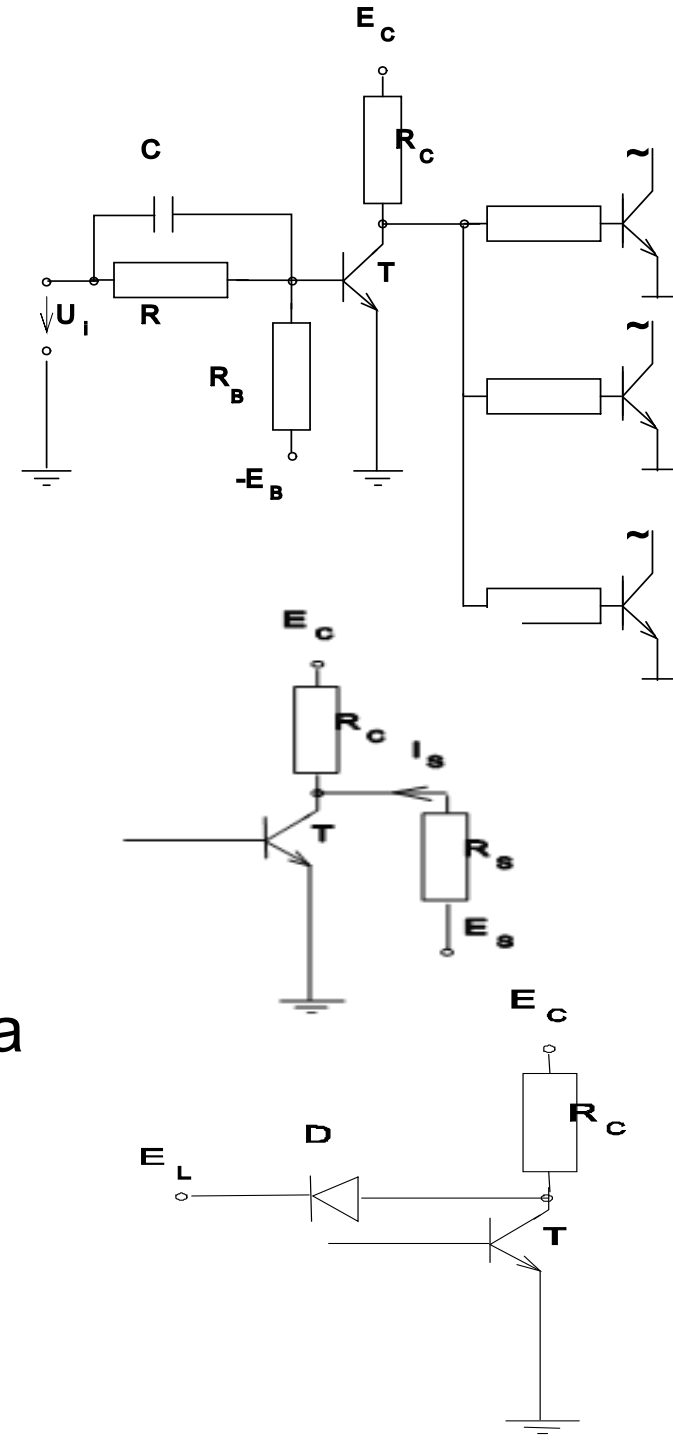
- Must be analysed, due to multiple possibilities to alter the values of circuit elements
- Circuit elements do not have the calculated (ideal) values, and a sum of 'bad' variations may drive the circuit in a wrong state, circuit behaviour may be affected
- Study Objectives:
 - Analysis of the influence generated by alteration of values for circuit elements
 - Getting the worst case, or the 'killing' combination of values
- Example for R_B , when transistor is off:
 - Transistor operates at maximum allowed temperature, making that collector-base current I_{CB} at off state having a maximum value I_{CBmax}
 - E_B presents minimum allowed value (90% normal value)
 - Tolerance for R_B is at the maximum allowed

All these 'mistakes' may generate a value for the calculated ratio lower than the chosen value for resistor R_B

- For R when transistor is saturated:
 - I_C has a maximum, given by a maximum allowed for E_C and a minimum value for R_C (within allowed range of tolerances)
 - β factors are at minimum β_{N0min}
 - Base power supply is at minimum E_{Bmin}

Load influence over circuit behaviour

- Driven gates (by circuit output) are considered as resistances R_s tied at E_s
- When T is off, U_{CEb} depends on R_s
- U_{CEb} influences calculus of R
- Load influence must be minimized
- Connect a diode D, with cathode tied to a limiting voltage E_L
- $V_{CEb} \approx E_L + V_D$ (not depending on load)



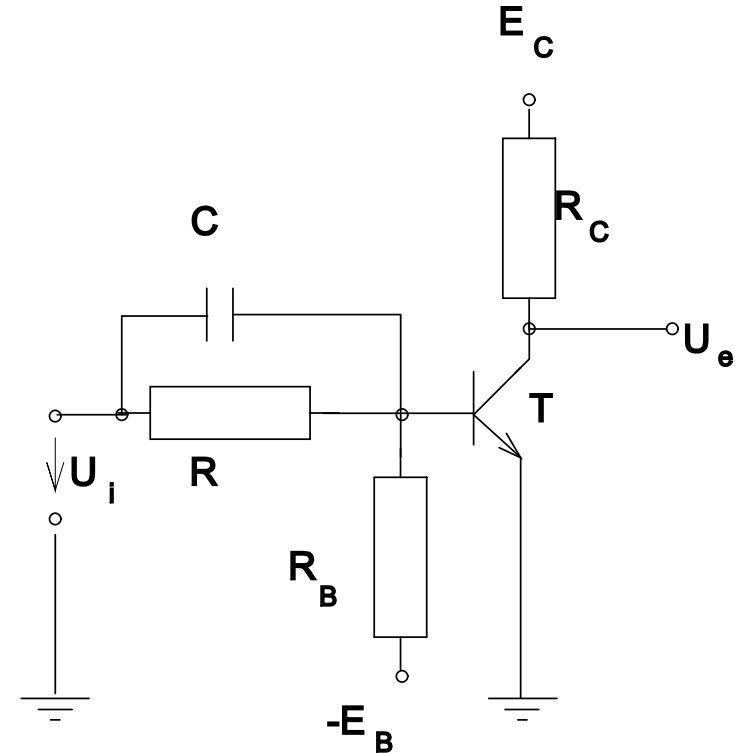
Dynamic regime behaviour

- Dynamic parameters of the inverter (switching times, their dependence on base currents)
 - For unblocking: $t_r = f(I_{Bd})$
 - For blocking: $t_c = f(I_{Bi})$ și $t_s = f(I_{Bi})$

Using relations:

$$R_B \leq \frac{E_B}{I_{C0}} = \frac{E_B}{I_{Bi}}$$

$$R \leq \frac{E_C - U_{BEs}}{\frac{E_C}{\beta R_C} + \frac{U_{BEs} + E_B}{R_B}} \approx \frac{E_C}{I_{Bd} + I_{Bi}}$$



Assumptions:

- Considering an input voltage swing from 0V to E_C
- Transistor becomes conducting (unblocked) in δt equal with t_r
- Transistor goes off in δt equal with $t_c + t_s$

Case of conduction:

$$\frac{du_c}{dt} \approx \frac{E_C}{\delta t} \quad I_c = C_1 \frac{du_c}{dt} = C_1 \frac{E_C}{t_{db}} \quad I_C = I_{Bd0} - I_{Bd}$$

Where I_{Bd0} represents base current for having over-driving:

$$I_{Bd0} = \frac{N_d I_{Cs}}{B_{N0}}$$

I_{Bd} represents normal direct base current at beginning of saturation

Results a first value for capacitance C:

$$C_1 = \frac{(I_{Bd0} - I_{Bd}) t_r}{E_C}$$

In a similar way will result the other value for C, given by the blocking regime:

$$C_2 = \frac{(I_{Bi0} - I_{Bi})(t_f + t_s)}{E_C}$$

A final value will be chosen:

$$C = \max(C_1, C_2)$$

2. For given layout, find out values for I_C and V_{CE} , having:
 $\beta=200$, $V_1=+12V$ and $V_2=5V$.

What will be if $V_2=0V$?

What will be if $V_2=12V$?

$$I_E = I_C + I_B \Rightarrow I_E = \beta I_B + I_B = (\beta + 1) I_B$$

$$V_2 - V_B - V_{BE} - V_E = 0$$

$$V_2 = I_B R_B + V_{BE} + (\beta + 1) I_B R_E$$

$$I_B = 0.05mA$$

$$I_C = \beta I_B = 10mA$$

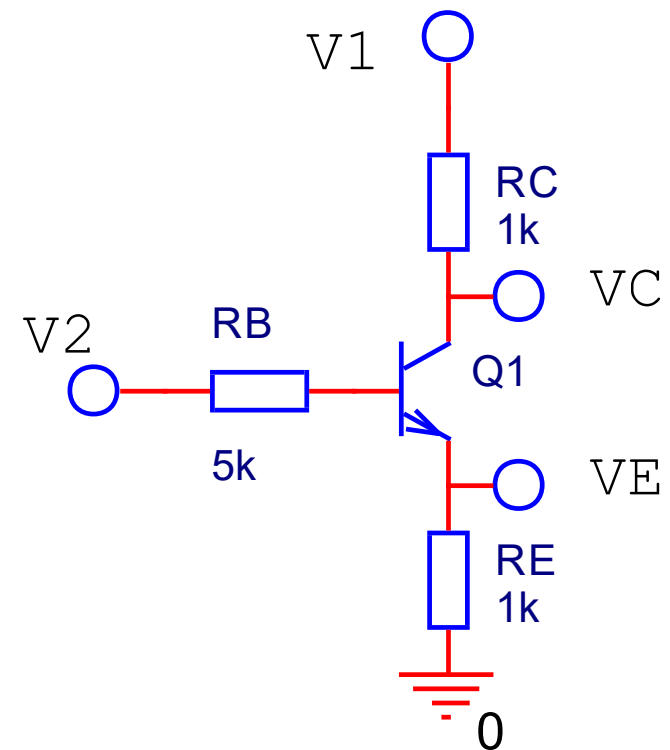
$$V_C = V_1 - I_C R_C = 2V$$

$$V_E = I_E R_E = 10V$$

- If $V_2=0V$ transistor will be off

$$I_C \approx I_E \approx 0$$

$$V_E = 0V, V_C = 12V, V_{CE} = 12V$$



- If $V_2=12V$ we have to determine if the transistor is in FAR or Saturation region

- We assume the transistor is in the FAR region

$$V_2 = I_B R_B + V_{BE} + (\beta + 1) I_B R_E$$

$$I_B = 0.05mA$$

$$I_C = \beta I_B = 10mA$$

$$V_{CE} = V_C - V_E$$

$$V_C = V_1 - I_C R_C = 2V$$

$$V_E = I_E R_E = 10V$$

- Thus the transistor is in Saturation

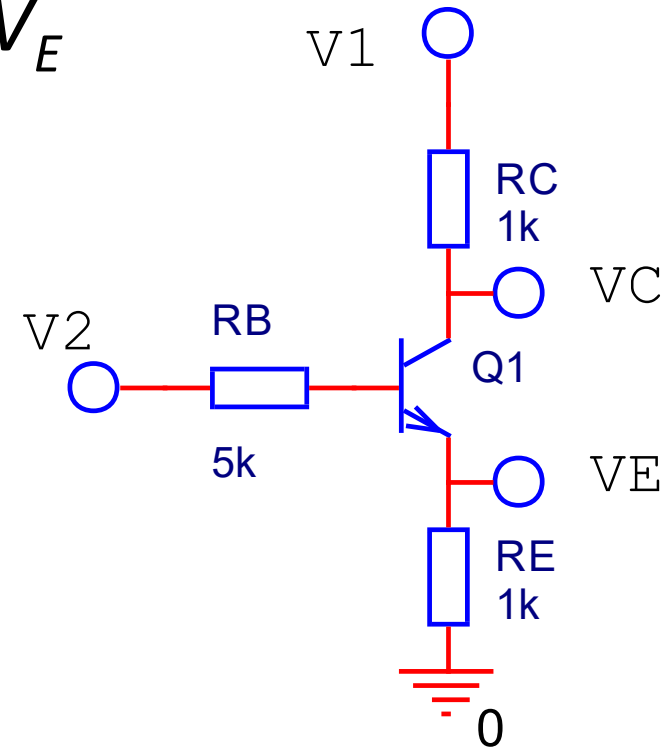
$$\begin{cases} V_1 = R_C I_{Csat} + V_{CEsat} + R_E (I_{Csat} + I_{Bsat}) \\ V_2 = R_B I_{Bsat} + V_{BEsat} + R_E (I_{Csat} + I_{Bsat}) \end{cases}$$

$$\begin{cases} V_1 = R_C I_{Csat} + V_{CEsat} + R_E (I_{Csat} + I_{Bsat}) \\ V_2 = R_B I_{Bsat} + V_{BEsat} + R_E (I_{Csat} + I_{Bsat}) \end{cases}$$

$$I_{Csat} = 5,3mA$$

$$I_{Bsat} = 1mA$$

$$V_{CEsat} = 0,2V$$



3. For this layout, determine I_D and V_{DS} , for: $V_T=4V$ and $K=400\mu A/V^2$.

$$V_G = \frac{R_{G2}}{R_{G1} + R_{G2}} V_{DD} = 10V$$

$$V_S = R_S I_D = R_S \frac{K}{2} (V_{GS} - V_T)^2$$

$$V_{GS} = V_G - V_S = V_G - R_S \frac{K}{2} (V_{GS} - V_T)^2$$

$$(V_{GS} - 4)^2 + (V_{GS} - 4) - 6 = 0$$

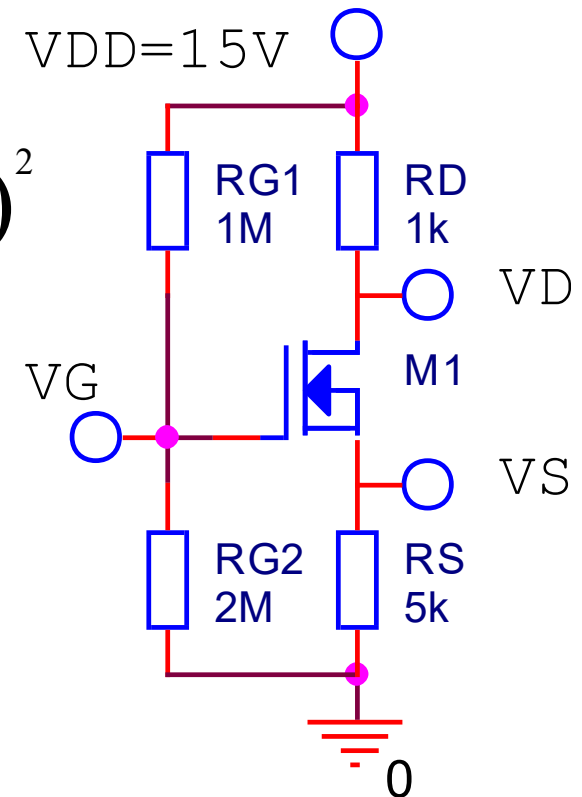
$$V_{GS} = 1, V_{GS} = 6$$

$$V_{GS} > V_T$$

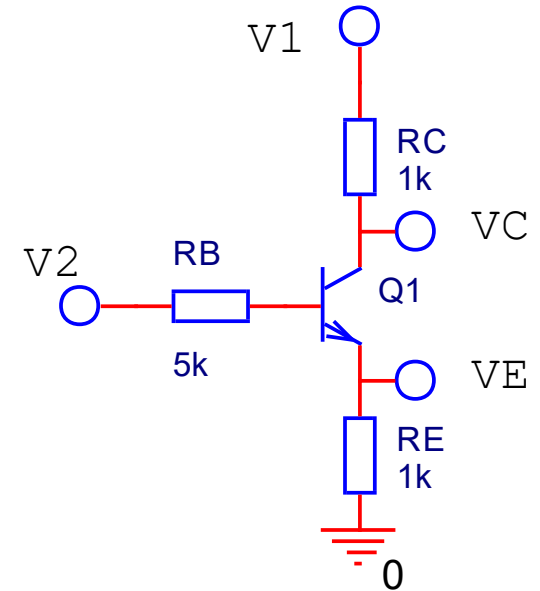
$$V_{GS} = 6$$

$$I_D = \frac{K}{2} (V_{GS} - V_T)^2 = 0,8mA$$

$$V_{DS} = V_{DD} - (R_D + R_S) I_D = 10,2V$$



4. For the given circuit, determine I_C and V_{CE} , knowing $\beta=100$, $V_1=12V$ and $V_2=10V$.



5. For the given circuit, determine I_D and V_{DS} , knowing $V_T=3V$ and $K=800\mu A/V^2$.

